

**TANIM: AN INTELLIGENT AGRICULTURAL FRAMEWORK
FOR CROP RECOMMENDATION AND SOIL HEALTH
MONITORING**

An Undergraduate Thesis

Presented to the
Faculty of Bachelor of Science in Computer Engineering
University of Science and Technology of Southern Philippines
Cagayan de Oro City

In Partial Fulfillment
of the Requirements for the Degree of
BACHELOR OF SCIENCE IN COMPUTER ENGINEERING

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October 2025

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Agriculture is important in the Philippines since it serves as the foundation of food security, livelihood, and rural development. Traditional soil management practices in the Philippines, such as those endorsed by the Department of Agriculture (DA), rely on generalized soil health monitoring and fertilizer recommendations using manual tools like the Minus-One Element Technique (MOET) and the Rapid Soil Test Device (RST). These approaches often fail to address field-specific variability, crop diversity, and seasonal dynamics, leading to suboptimal resource utilization, as reported by the Department of Agriculture Region 10 (2023). Historically, soil testing requires farmers to transport soil samples to nearby laboratories located at Department of Agriculture offices, a process that is often expensive, time-consuming, and inaccessible in remote areas, as highlighted by Pagaduan, J. L. (2025).

In relation to this, the Department of Agriculture – Regional Field Office 10 (DA–RFO 10 or Aggie 10) conducted a soil sampling drive for rice and corn production areas in the municipalities of Sugbongcogon, Binuangan,

Magsaysay, and Gingoog City from March 26–28, 2025. Aggie 10 aimed to complete soil sampling and testing in priority sites across Region 10 by May 2025 to align with the wet cropping calendar. Despite these initiatives, farmers still largely depend on traditional and observational methods. The study by Samaniego, A. L. and Gallego, M. (2024) reported that only 1.03% of farmers use soil-test kits and another 1.03% utilize government laboratory services, while the remaining 97.94% rely on ocular observation or “eyeballing” of soil and crop conditions. Furthermore, crop selection decisions are often based on historical practices rather than data-driven insights, limiting farmers’ ability to adapt to changing soil and climate conditions (Sarma et al., 2024). With advancements in technology such as the Internet of Things (IoT), there is significant potential to improve crop yields and reduce agricultural losses.

Soil health is a fundamental pillar of sustainable agriculture, underpinning crop productivity, food security, and environmental resilience. According to Lal, R. (2024), degraded soils characterized by nutrient deficiencies, imbalanced pH levels, or reduced organic matter lead to lower yields, increased susceptibility to pests, and economic losses for farmers. In Region 10 (Northern Mindanao, Philippines), a major agricultural hub producing rice, corn, banana, and cacao, maintaining optimal levels of nitrogen (N), phosphorus (P), potassium (K), and a soil pH range of 6.0–7.5 is essential for nutrient

availability and healthy crop growth, as discussed by Morgan and Connolly (2013) and Sondhiya and Singh (2024). Data from the Department of Agriculture Region 10 (2024) indicate that acidic soils in Bukidnon restrict phosphorus uptake, while nitrogen deficiencies in rice fields can reduce yields by up to 20%. Moreover, proper crop selection based on soil condition and season, combined with precise fertilizer application, is crucial to maximizing productivity and minimizing environmental impacts such as nutrient runoff (Patil et al., 2023).

Globally, agricultural losses caused by soil degradation, inefficient nutrient management, and climate variability pose a serious threat to food security and economic stability. The Food and Agriculture Organization (FAO, 2023) estimates that approximately 33% of the world's soils are moderately to highly degraded, resulting in annual crop yield losses of up to 25% in some regions. The Economics of Land Degradation (ELD) initiative (2015) further reports that land degradation leads to global economic losses of up to USD 10.6 trillion per year, equivalent to nearly 17% of global Gross Domestic Product (GDP), due to reduced agricultural productivity, ecosystem service loss, and increased vulnerability to climate extremes. These losses are exacerbated by improper fertilizer use, where over-application leads to nutrient runoff and environmental damage, while under-application results in nutrient deficiencies and yield reduction.

In the Philippine context, these global challenges are strongly reflected at the national and regional levels. The Department of Agriculture (2024) reported that approximately 80% of agricultural soils in the Philippines are deteriorated, significantly limiting crop productivity and fertilizer efficiency. Climate-related disturbances such as typhoons, droughts, and flooding further contribute to agricultural losses, with the Department of Agriculture estimating P785 million in crop damage in late 2024 alone due to successive weather events. In major agricultural regions such as Northern Mindanao (Region 10), nutrient-depleted and acidic soils, combined with non-site-specific fertilizer practices, contribute to yield reductions in staple crops such as rice and corn, increasing farmers' production costs and reducing income stability. These persistent losses underscore the urgent need for data-driven, location-specific, and climate-aware agricultural decision-support systems.

Advancements in Internet of Things (IoT) technologies and machine learning (ML) offer promising solutions for precision agriculture. IoT-based sensors, including NPK, pH, moisture, and temperature probes, enable continuous soil health monitoring, while machine learning algorithms analyze large and complex datasets to generate data-driven insights and recommendations (Kumar et al., 2024 & Lavanya et al., 2024). Modern farming systems leverage these technologies to optimize crop production, improve fertilizer efficiency,

and reduce waste, ultimately enhancing yield and sustainability (Elbasi et al., 2023). By integrating Internet of Things (IoT) sensor data with historical records, and agricultural zoning data, intelligent systems can provide crop recommendations tailored to the agricultural conditions of Region 10.

This study proposes a Soil Health Monitoring and Recommendation System that leverages IoT sensors and machine learning techniques to provide recommendations for crop planting and fertilizer application in Bukidnon. Targeting key crops such as Ampalaya, Beans (Soybean), Cassava, Corn, Eggplant, Sorghum, Squash, Sweet Pepper, Tobacco, and Tomato, the system integrates soil health parameters (NPK, pH, moisture, humidity, and temperature), soil and crop data from the Department of Agriculture - Region 10, and additional online agricultural resources to predict suitable crops and recommend fertilizer application. By addressing field-specific variability and seasonal dynamics, the proposed system aims to enhance agricultural productivity, reduce resource waste, and promote sustainable farming practices. Aligned with the Sustainable Development Goals (SDGs) 2 (Zero Hunger), 12 (Responsible Consumption and Production), and 15 (Life on Land), this research contributes to food security, environmental conservation, and sustainable agricultural development in the Philippines.

1.2 Statement of the Problem

Traditional soil management and crop selection practices in Region 10 remain largely generalized and manual, limiting their effectiveness in addressing field-specific agricultural conditions. Farmers primarily rely on standard guidelines and conventional assessment tools provided by the Department of Agriculture, which do not sufficiently consider variations in soil properties, crop requirements, and seasonal environmental changes. This lack of precision often results in inefficient fertilizer application, suboptimal crop selection, reduced yields, increased production costs, and gradual soil degradation (Lal, R., 2023). These issues are particularly evident in Bukidnon, where diverse crops such as Ampalaya, Beans (Soybean), Cassava, Corn, Eggplant, Sorghum, Squash, Sweet Pepper, Tobacco, and Tomato are cultivated under varying soil conditions.

Despite the importance of soil health information in agricultural decision-making, several problems persist that hinder the effective use of technology and data-driven approaches in the region. These problems include:

1. Inconsistent and limited availability of accurate, real-time soil health data (pH, NPK, salinity, moisture, and temperature) across different farming zones in Bukidnon, making it difficult to assess actual field conditions.

2. The absence of an integrated decision-support mechanism that analyzes soil health data, resulting in unreliable or generalized crop suitability and fertilizer recommendation application.
3. Limited access to user-friendly digital platforms that can translate complex agricultural data into clear, practical recommendations understandable to farmers in Bukidnon.
4. Uncertainty regarding the effectiveness and scalability of existing IoT-based agricultural monitoring systems in supporting sustainable farming practices across multiple farms and locations in Bukidnon.

1.3 Objectives of the Study

The main objective of this study is to develop an IoT-based crop recommender system integrated with machine learning for crop recommendation based on soil health parameters (pH, NPK, salinity, moisture, and temperature) for farmers, specifically the following:

1. Implement a machine learning model to predict suitable crops and recommend fertilizers application for farming practices in Bukidnon based on soil health data.
2. Develop a mobile application to monitor soil health, crop predictions, and fertilizer schedules for the farmers.

3. Develop a website application to monitor soil health insights for different zoning of farms, farmers data and information on their farming timeline.
4. Evaluate the IoT-based crop recommender system's accuracy and usability in agricultural settings in Bukidnon.

1.4 Significance of the Study

The significance of this study lies in its potential to transform agricultural practices in Bukidnon by developing an integrated Internet of Things (IoT) and machine learning-based system for soil health monitoring, crop recommendation, and recommend fertilizer application. Traditional soil testing methods provided by the Department of Agriculture are often underutilized by farmers due to inaccessibility and time-consuming procedures. As a result, many farmers rely on ocular observations, which may lead to poor crop selection and increased production risks (Samaniego, A. L. & Gallego, M., 2024). This research addresses critical limitations of traditional soil management practices, particularly the reliance on generalized recommendations and the lack of zone-specific insights, which frequently result in inefficient resource utilization.

By providing precise soil health data and data-driven crop and fertilizer recommendations, the proposed system enables farmers to optimize crop yields

while reducing unnecessary fertilizer expenses. Furthermore, the system will be able to provide the Department of Agriculture a Geographic Information System (GIS) zoning maps for farms across Bukidnon for better monitoring and supports sustainable farming practices aligned with environmental conservation goals. The study contributes to the advancement of precision agriculture by offering a scalable IoT and machine learning framework that promotes data-driven decision-making in farming. Ultimately, this research has the potential to improve agricultural productivity, environmental sustainability, and food security in Bukidnon, in support of Sustainable Development Goals (SDGs) 2 (Zero Hunger), 12 (Responsible Consumption and Production), and 15 (Life on Land).

Farmers. The system will significantly assist farmers by providing accessible soil health monitoring data, data-driven crop recommendations, and optimized fertilizer usage. This enables farmers to reduce production risks, improve decision-making, and maximize crop yields while minimizing resource waste.

Department of Agriculture. The study will support the Department of Agriculture in Region 10 by offering timely soil health insights that can improve the accuracy of fertilizer distribution programs and enhance agricultural zoning strategies. The system may also serve as a tool for planning

interventions and monitoring soil health trends to promote sustainable agricultural development.

Agriculturists. This research benefits agriculturists by serving as a reliable reference for assessing soil conditions and crop suitability. It enhances their ability to provide evidence-based recommendations to farmers, reducing dependence on manual and generalized assessment methods.

Future Researchers. The study will serve as a foundation for future research in smart farming and precision agriculture. It opens opportunities for further development, such as integrating remote sensing data, advanced machine learning models, and different agricultural techniques. Future researchers may also adapt and expand the system to other regions or crop categories, making it a scalable model for broader agricultural applications.

1.5 Scope and Limitations

This study focuses on agricultural areas located in Budkinon, Philippines, where selected crops—namely Ampalaya, Beans (Soybean), Cassava, Corn, Eggplant, Sorghum, Squash, Sweet Pepper, Tobacco, and Tomato—are commonly cultivated. These crops are the primary focus of the system’s crop recommendation and fertilizer application process. The system is designed to analyze soil health parameters including (1) nitrogen, (2) phosphorus, (3)

potassium, (4) soil moisture, (5) temperature, (6) pH level, and (7) soil salinity. These parameters are combined with the soil and crop data from the Department of Agriculture - Region 10, forming the basis for crop recommendation and fertilizer management.

The study is geographically limited to selected farms across Bukidnon, ensuring relevance to local soil health characteristics. The system emphasizes the integration of IoT-based soil sensor and machine learning techniques for data analysis and recommendation generation. However, the soil sensor used in this study is limited to seven (7) soil parameters and does not include micronutrients or any soil organic matter. Additionally, real-time continuous soil monitoring is not implemented; instead, the system relies on periodic data collection and analysis. These limitations may affect the granularity of recommendations but are sufficient to demonstrate the feasibility and effectiveness of the proposed system within the defined scope.

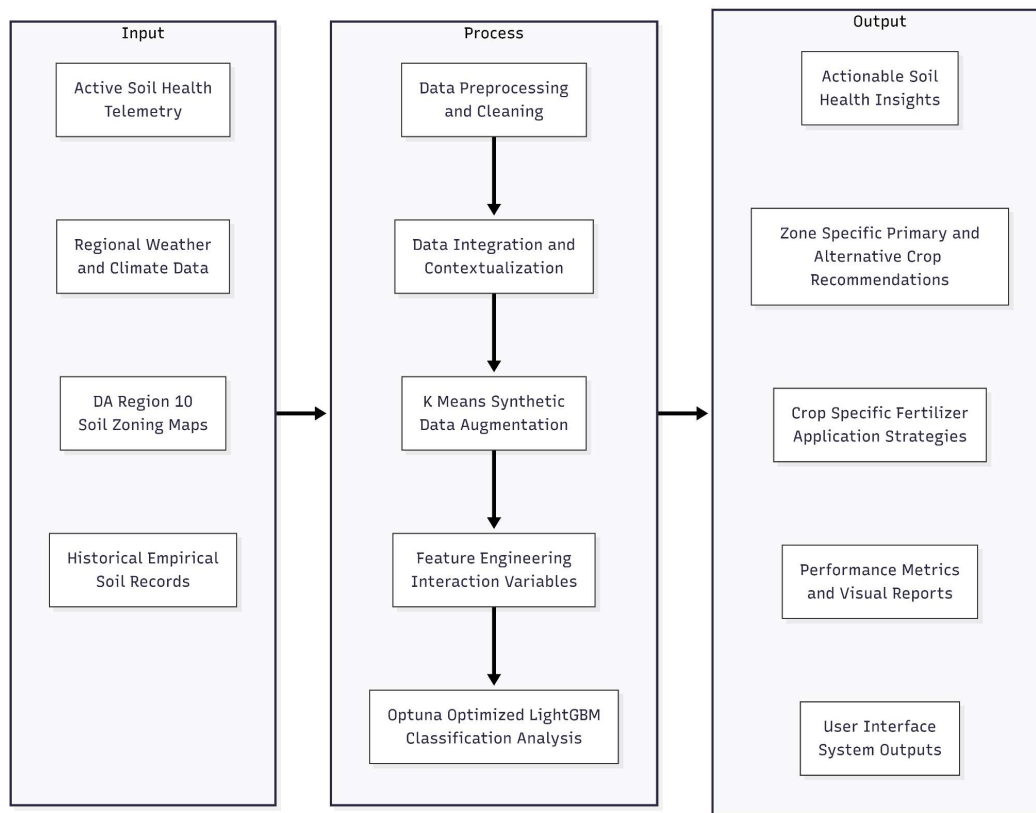
1.5.1 Conceptual Framework

In this section, the study's conceptual framework is presented through an Input-Process-Output (IPO) model, which outlines the structure of the soil health monitoring, crop and fertilizer recommendation system. The framework integrates multiple components, each playing a vital role in developing an

efficient, data-driven solution for sustainable agriculture.

Figure 1.1

IPO Model



The TANIM system operates upon an Input Process Output model explicitly engineered to generate localized crop and fertilizer recommendations for farmers across Region 10, Philippines. The Input phase captures active soil health telemetry, such as Organic Matter serving as a Nitrogen proxy, Phosphorus, Potassium, soil pH, moisture, salinity, and temperature, gathered continuously via an ESP32 microcontroller and its attached Internet

of Things sensor network. To establish a comprehensive agricultural baseline, these localized sensor inputs run parallel to regional weather and climate data, Department of Agriculture Region 10 soil zoning maps, and historical empirical soil records.

Transitioning into the Process layer, these raw inputs undergo a rigorous and multistage computational pipeline. The acquired telemetry is first subjected to programmatic data preprocessing and cleaning protocols, systematically removing empty reads and sensor anomalies to guarantee absolute diagnostic consistency. Within the data integration and contextualization phase, severe historical imbalances present in the agricultural dataset are actively neutralized via K Means synthetic data augmentation, ensuring all crop classes hold equal representational weight. Furthermore, a specialized feature engineering protocol derives crucial agricultural interaction variables, such as pH controlled nutrient availability, dynamically from the raw data. This deeply contextualized and balanced matrix is subsequently evaluated by an Optuna optimized Light Gradient Boosting Machine framework, which analyzes complex soil condition patterns to precisely isolate and predict biological crop suitability.

Governed by this classification analysis, the Output phase is generated. The system delivers actionable soil health insights alongside high confidence

and zone specific primary and alternative crop recommendations. These predictive classifications bridge seamlessly into secondary subsystems that formulate appropriate and crop specific fertilizer application strategies. The operational validity of the entire pipeline is continuously tracked via rigorous performance metrics, including analytical confusion matrices, prediction error monitoring, and F1 score dashboards, which are compiled natively into visual reports and delivered through an accessible user interface. Ultimately, this seamless and data driven cycle equips agricultural operators with the empirical insight required to optimize resource utilization, maximize productivity, and enforce sustainable farming practices.

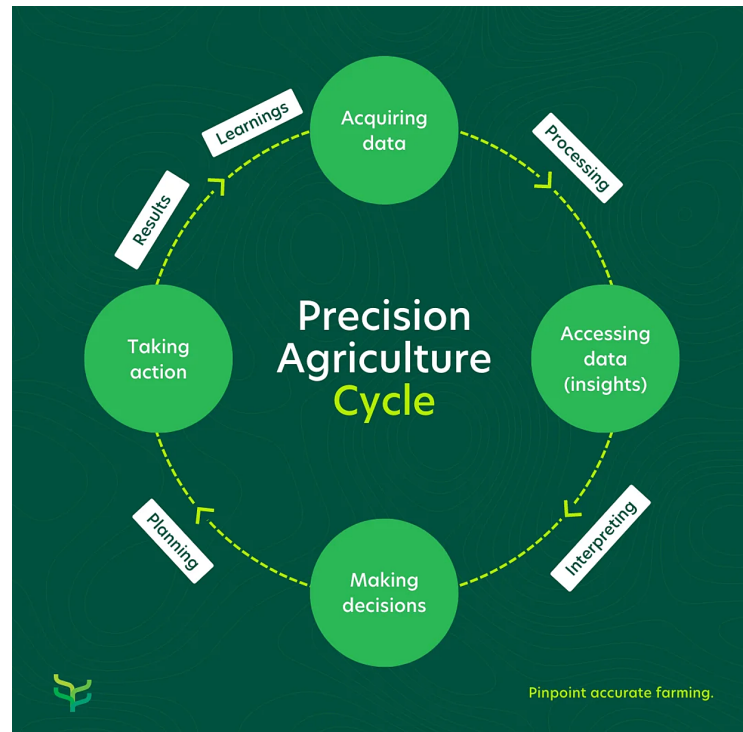
1.5.2 Theoretical Framework

This study utilized Precision Agriculture Theory, a strategy that gathers, processes data driven insights to optimize farming practices based on conditions rather than relying on generalized farming practices recommendations, which is supported by the study of Pierre C. Robert states that an integrated farming approach designed to enhance long-term production efficiency, productivity, and profitability while minimizing environmental impacts. It involves using technology and data to optimize farming practices at a site-specific level. The theory highlights the importance of applying the right input, at the right

place, at the right time, and in the right amount to maximize productivity while conserving resources.

With the integration of IoT sensors which will provide measurements of soil health parameters the precision agriculture theory will be fully operational. The soil parameters will serve as a foundation for the Multi-Criteria Decision Analysis methods to generate actionable recommendations for farmers. Moreover, machine learning boosting algorithms will be used to analyze soil and environmental patterns to predict crop suitability and yield potential.

Figure 2 explains the Precision Agriculture Cycle on how farming decisions become data-driven. It starts with acquiring data through IoT sensors and other tools that measure soil and environmental conditions. This data is then processed into insights to understand soil fertility and crop needs. Using these insights, decisions are made with the help of Machine Learning (ML) for prediction and Multi-Criteria Decision Analysis (MCDA) for evaluating multiple factors like soil parameters, climate, and zoning. The next step is taking action, where farmers apply precise fertilizers, choose suitable crops, or adjust irrigation. Finally, the results and learnings feed back into the system, ensuring continuous improvement in productivity, efficiency, and sustainability.

Figure 1.2*Precision Agriculture Cycle*

By combining Machine Learning’s predictive capability with MCDA’s structured decision making process, the study applies the principles of precision agriculture in a holistic manner. This ensures that recommendations for crop selection and fertilizer application are data-driven, addressing the variability of soils in Bukidnon. Thus, Precision Agriculture Theory serves as the guiding lens that connects IoT-based data collection and machine learning prediction to support decision-making, ultimately promoting sustainable and efficient farming practices.

1.6 Definition of Terms

This study defined the following terms that are mentioned in this study.

Agriculture The science, art, and practice of cultivating plants and livestock, serving as the foundation for food security, livelihoods, and rural development in regions like the Philippines.

Agricultural Zoning The practice of designating specific areas for agricultural use to conserve farmland, prevent urban conflicts, and promote orderly rural growth, often including minimum lot sizes and restrictions on non-farm activities.

Analytic Hierarchy Process (AHP) A structured decision-making technique that organizes complex problems into a hierarchy of goals, criteria, and alternatives, using pairwise comparisons to derive priorities and rank options.

Boosting Algorithms A family of machine learning techniques that convert weak learners into strong ones by iteratively training models to correct errors from previous ones, commonly used in regression and classification tasks.

Crops Plants or plant products grown and harvested extensively for profit

or subsistence, categorized into food, feed, fiber, oil, ornamental, and industrial types based on their uses.

Fertilizer A natural or synthetic substance added to soil to supply essential nutrients like nitrogen, phosphorus, and potassium, enhancing plant growth and productivity in agriculture.

Internet of Things (IoT) in Agriculture The integration of connected sensors, devices, and networks in farming to monitor soil, weather, crops, and livestock, enabling data-driven decisions for improved efficiency and sustainability.

Machine Learning A subset of artificial intelligence that enables computers to learn from data and improve performance on tasks without explicit programming, often using algorithms to identify patterns and make predictions.

Multi-cropping The practice of growing two or more crops on the same piece of land in a single year, either sequentially or simultaneously, to enhance productivity, diversify income, and improve soil health.

NPK The abbreviation for the three primary macronutrients in soil and fertilizers: nitrogen (N), phosphorus (P), and potassium (K), essential for plant growth, development, and overall health.

Precision Agriculture A farming management strategy that uses data from sensors, GPS, and analytics to optimize inputs like water and fertilizers, responding to variability in crops for improved efficiency and sustainability.

Soil Health The continued capacity of soil to function as a vital living ecosystem that sustains plants, animals, and humans, encompassing biological, physical, and chemical properties.

Soil pH A measure of soil acidity or alkalinity on a scale from 0 to 14, indicating hydrogen ion concentration, which influences nutrient availability and plant growth.

Soil Salinity The accumulation of soluble salts in soil that can impair plant growth, often measured by electrical conductivity, with levels above 4 dS/m considered saline.

Sustainable Agriculture Farming practices that meet current food needs while preserving resources for future generations, balancing environmental health, economic viability, and social equity.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents some preliminary concepts and known results that are needed in this study.

2.1 Soil Health in Northern Mindanao

The literature underscores the critical role of soil health in sustainable agriculture and the limitations of traditional methods. Lal (2024) emphasizes soil degradation's impact on global food security, while Morgan and Connolly (2013) detail nutrient uptake mechanisms and challenges in imbalanced soils. In the Philippine context, Department of Agriculture Region X (2023) highlights regional soil issues, such as acidity in Bukidnon and nutrient shortages in rice fields, alongside generalized zoning maps. According to the study of Arlyn and Carmelito (2024), which assessed soil fertility in irrigated lowland areas across Northern Mindanao, revealing strongly to moderately acidic soils (pH 4.5-6.0) with variable organic matter content, peaking in Bukidnon at up to 3.5%. The study employed soil sampling and laboratory analysis to monitor key parameters like pH, and NPK, highlighting deficiencies in phosphorus and potassium that limit rice yields. This underscores the need for regular soil

health monitoring to track degradation trends and inform site-specific interventions in Region 10's rice-dominated landscapes; this is supported by the study of ? which uses several factors of crops and uses those inputs for random forest algorithms.

Crop production trends in Northern Mindanao, note that fertilizer inefficiencies in Bukidnon and Misamis Oriental contribute to yield gaps in corn and rice, with average applications exceeding recommendations by 15-20% based on the study of Canatoy (2018). The review recommends precision fertilizer strategies, such as variable-rate applications, to optimize nutrient use efficiency and minimize environmental impacts in Region 10's diverse agroecosystems. On the other hand, Calalang et al. (2014) reviewed soils and crop suitability in Bukidnon Highlands, highlighting how volcanic-derived soils support high-value crops like pineapple and coffee, but require lime amendments for acidity management to sustain productivity. The study links soil properties to crop performance, advocating for diversified cropping systems to enhance resilience in Northern Mindanao.

2.1.1 Zoning in Agriculture

In agriculture, zoning is the process of dividing land into management areas based on topographic features, soil fertility, and climate—all of which have

a direct impact on crop suitability. Zoning has been used in several areas of the Philippines to maximize land use and improve site-specific agricultural recommendations. Watson et al. (2025) used the Food and Agriculture Organization of the United Nations (FAO) land evaluation to characterize and categorize soil series in Ilocos Norte and perform crop suitability assessments for rice, maize, garlic, onion, and tomato. According to their research, morphological characteristics and soil physicochemical characteristics are important factors in determining whether an area is considered highly suitable (S1), moderately suitable (S2), or marginally suitable (S3).

The use of zoning for particular crops is further illustrated by regional studies. In Samar Province, Poliquit et al. (2019a) evaluated crop suitability and soil fertility, identifying nutrient deficiencies like low levels of organic matter, phosphorus, and nitrogen as productivity-limiting factors. The significance of mapping local soil fertility for focused input management was highlighted by their findings. Similarly, Bato (2018) created a Geographic Information System (GIS) based banana cultivation suitability map that illustrates the spatial variations in suitability across the Philippine landscape. Slope, elevation, rainfall, and soil properties were all integrated in the study to create a visual guide that farmers and policymakers could use to identify the best places to grow bananas.

These zoning studies together make a strong case for creating smart frameworks that combine soil and crop suitability with cutting-edge technologies to give site-specific crop and fertilizer recommendations.

2.2 Machine Learning

Machine learning (ML) is becoming very useful in agriculture, it enables researchers and farmers to process huge amounts of data. It provides predictive insights related to soil health, crop recommendation, fertilizer and so on. Unlike traditional methods that rely on generalized formulas or manual observations, ML can accommodate complex and internal relationships between the soil, climate and crop parameters. This will enable them to make site-specific recommendations which will increase productivity, reduce costs and improve sustainable farming practices.

As noted by Islam et al. (2023), ML enabled systems are flexible. They not only analyze nutrient levels of soil but also suggest crops that can be grown in various environments. They do this with more accuracy and precision than manual assessments. According to Parganiha and Verma (2024), optimized ML models like LightGBM can yield predictive accuracies as high as 97% for crop yield forecasting. These models are easily scalable for larger agricultural zones like Northern Mindanao.

2.2.1 Random Forest

Random Forest (RF) is an ensemble algorithm that constructs multiple decision trees and aggregates their outputs through majority voting (classification) or averaging (regression). Incorporating bootstrapped data subsets and randomly selected features minimizes overfitting and enhances the RF model's resilience to the inaccurate data typical of many agricultural applications. This is useful in examining soils, where data quality is often compromised due to sparse sampling and other uncontrolled environmental factors.

In predicting fertilizer amounts and estimating crop inputs, ? demonstrated its capability in capturing nonlinear relationships integrated among soil parameters and environmental factors. The study found that RF outperformed linear models in terms of accuracy, showing its adaptability across diverse farming conditions. Islam et al. (2023) integrated RF into a crop recommendation framework, successfully suggesting suitable crops based on soil nutrient levels and weather data.

2.2.2 Boosting Algorithms

With the current technologies, Islam et al. (2023) developed an ML-enabled system for soil nutrient monitoring, and crop recommendations, evaluated through field experiments to enhance productivity and sustainability.

This approach supports multi-crop applications relevant to Region 10. Parganiha and Verma (2024) proposed an optimized LightGBM model for soil analysis and crop yield prediction, incorporating data preprocessing, feature selection, and hyperparameter tuning, achieving up to 97% predictive accuracy. LightGBM’s efficiency, as noted by Data Overload (2024), makes it ideal for large-scale agricultural tasks in Region 10.

Table 1: Parameters for Different Gradient Boosting Algorithms (*Salvador (2024)*)

Algorithm	Parameter	Value
Adaptive Boosting (AdaBoost)	learning_rate	0.5
	n_estimators	200
Cat Boosting (CatBoost)	depth	4
	iterations	300
	iterations	300
Gradient Boosting Machine (GBM)	learning_rate	0.3
	max_depth	3
	n_estimators	300
Light Gradient Boosting Machine (LightGBM)	learning_rate	0.1
	n_estimators	200
	num_leaves	31
Extreme Gradient Boosting (XGBoost)	learning_rate	0.3
	max_depth	3
	n_estimators	300

Table 1 outlines the hyperparameters used for training the different gradient boosting algorithms considered in this study. Each model was configured

with commonly used parameter settings based on literature benchmarks and preliminary tuning to balance performance and computational efficiency. For AdaBoost, the number of estimators was set to 200 with a relatively high learning rate of 0.5 to speed up convergence. CatBoost employed a maximum tree depth of 4, 300 boosting iterations, and a learning rate of 0.3. This reflects its ability to handle categorical features effectively. The Gradient Boosting Machine was configured with a learning rate of 0.3 and a tree depth of 3, alongside 300 estimators to control complexity while maintaining accuracy. LightGBM used a lower learning rate of 0.1 with 200 estimators and 31 leaves, capitalizing on its efficiency in handling large datasets. Lastly, XGBoost was set with 300 estimators, a learning rate of 0.3, and a maximum depth of 3, representing a balanced configuration widely adopted in classification tasks.

2.2.3 Decision Tree Analysis

Decision tree analysis is a supervised machine learning approach that uses a tree-like model of decisions and their possible consequences to classify or predict outcomes. It is widely applied in agriculture because of its simplicity, interpretability, and ability to handle both categorical and numerical data. By splitting data into smaller subsets based on key attributes such as soil nutrients, moisture, pH, and climatic conditions, decision trees can provide

farmers with clear and actionable recommendations.

A study of Lavanya et al. (2024) employed Gradient Boosting Decision Trees for multi-crop fertilizer recommendation, integrating soil and climate data, and reported 95% accuracy. This highlights the method's strength in managing Region 10's diverse cropping systems where multiple environmental factors interact.

Similarly, Sudha and Hooda (2022). examined the applicability of decision tree algorithms in agriculture for classification tasks, emphasizing their effectiveness in handling diverse datasets related to crop type, soil fertility, and environmental conditions. Their study concluded that decision trees not only enhance prediction accuracy but also improve the interpretability of agricultural data, enabling researchers and farmers to make evidence-based decisions. This demonstrates that decision tree-based models are reliable tools for optimizing fertilizer management, predicting yields, and supporting sustainable farming practices.

2.3 Crop and Fertilizer Recommendation Systems

In the Philippines, crop and fertilizer recommendations remain very general and imprecise. According to Samaniego and Gallego (2024), it was found that as much as 97.94% of farmers rely on ocular observations or visual

inspection of plants and soil, while only 1.03% use soil-test kits or government laboratories. This reliance on manual methods ends up either over or under-application of fertilizer, contributing to low yields and unnecessary costs. Farmers typically follow broad guidelines or personal judgement in fertilizer management, but these approaches lack the precision required to account for variations in soil health, nutrient level, and crop demands.

To address these inefficiencies, several studies have explored machine learning and data-driven approaches for fertilizer optimization. Sani et al. (2023) employed Random Forest algorithms to predict fertilizer requirements with various soil and crop inputs, achieving high accuracy. Lavanya et al. (2024) demonstrated Gradient Boosting Decision Trees (GBDT) could yield multi-crop fertilizer recommendations with an accuracy of 95% by integrating the soil and climate data. Canatoy (2018) also recommended precision fertilizers, like using different amounts of fertilizers based on location, to avoid nutrient waste and environmental impact. These findings demonstrate the potential of ML models to provide more efficient and tailored fertilizer recommendations compared to traditional methods.

Crop recommendation systems are equally important for optimizing agricultural practices. Islam et al. (2023) developed a machine learning enabled soil monitoring and crop recommendation system that uses an an-

alyzes soil nutrient levels and environmental conditions to suggest suitable crops across multiple environments. The study highlighted the adaptability of ML in multi-cropping systems, showing that intelligent algorithms can go beyond input optimization to guide farmers in selecting crops that align with specific soil and climatic conditions.

2.4 Soil Health Sensor in Agriculture

Soil sensors play a critical role in modern agriculture as they provide accurate and real-time data about soil conditions. By measuring key parameters such as soil moisture, temperature, pH, salinity, and nutrient levels, these sensors help farmers assess overall soil health and make informed decisions for crop management. This is supported by the study of Yin et al. (2021), in which they highlighted the four aspects that will be the main focuses for future soil sensing. First is to improve the sensing performance (e.g., sensitivity and specificity) and reliability for key soil parameters, with little interference from background noise; Second is to develop a sufficient low-power consumption Wireless Sensor Network (WSN) with powerful data processing and long-range wireless communication capability; Third is to develop versatile soil sensing platforms that can be distributed in large-scale to collect real-time soil microenvironment data continuously; and the Fourth is to develop self-powered

or power-independent sensors and sensing platforms that are low-cost, reliable, and maintenance free.

The use of soil health sensors enables precision agriculture, where inputs like water, fertilizers, and pesticides can be applied more efficiently, reducing waste and minimizing environmental impact. In addition, continuous monitoring of soil health through sensors helps in early detection of soil degradation issues such as nutrient depletion, salinization, or compaction. This allows for timely interventions that improve soil productivity and sustainability. The integration of soil sensors with Internet of Things (IoT) platforms further enhances their effectiveness by enabling wireless data transmission, cloud-based analytics, and automated decision-making systems for smart farming.

Numerous studies in the field of agriculture integrate modern advancements like Internet of Things (IoT) , it is supported by the study of Sondhiya and Singh (2024), which explores Internet of Things (IoT) and Machine Learning (ML) for soil monitoring in Zea mays (corn) in Kibangay, Lantapan, Bukidnon, demonstrating improved nutrient management. The study utilized NPK and pH sensors with an ESP32 microcontroller and LoRaWAN, achieving 95% agreement with the Department of Agriculture’s Soil Test Kit. However, its single-crop focus limits applicability to Region 10’s diverse crops like rice, banana, and cacao. Moreover, the study of Kumar et al. (2024a) developed an

IoT-based soil monitoring system using Arduino-based NPK and moisture sensors for rice and maize, achieving 95% nutrient detection accuracy, validating the potential of IoT for monitoring.

2.5 Synthesis

The reviewed studies provide comprehensive insights into soil health, crop productivity, and machine learning applications in agriculture. Research gained in-depth understanding of the soil health challenges in Northern Mindanao, particularly with issues like soil acidity in Bukidnon and nutrient deficiencies in rice fields. Regional assessments, including those from the Department of Agriculture Region 10 (2023, 2024) and the work of Temario and Lapoot (2024), confirmed the variability of soil properties and their direct impact on yield performance. These findings are consistent with earlier insights from Canatoy (2018), who noted the inefficiencies in fertilizer applications, and from Calalang et al. (2014), who linked volcanic-derived soils to both opportunities and limitations for high-value crops. Together, these studies stress the importance of site-specific soil health monitoring and precision interventions to counteract degradation and sustain agricultural productivity. However, most of these studies remain limited to general recommendations or crop-specific assessments, leaving a gap in developing integrated, data-driven

systems tailored to multiple crops in Northern Mindanao.

Zoning has also emerged as a crucial strategy for sustainable land use. Efforts by Watson et al. (2025) using FAO land evaluation in Ilocos Norte, the crop suitability mapping in Samar by Poliquit et al. (2019b), and the GIS-based banana suitability mapping by Bato (2018) demonstrated how soil fertility, topography, and climate can be translated into actionable agricultural zoning. These initiatives underscore the value of integrating spatial data with soil and crop requirements. Yet, these zoning studies are often region-specific and crop-focused, lacking the integration with modern technologies such as machine learning and IoT sensors that could make zoning more dynamic and adaptive to current farming needs.

The incorporation of machine learning adds a new dimension to these agricultural practices. Random Forest applications by ? and Islam et al. (2023) showed how nonlinear soil and environmental data can be used to generate accurate fertilizer and crop recommendations, while boosting algorithms such as LightGBM, as explored by Parganiha and Verma (2024), achieved high predictive accuracies in crop yield forecasting. Complementary approaches like decision tree analysis, highlighted in the work of Lavanya et al. (2024) and Sudha and Hooda (2022), further support the adaptability and interpretability of ML models. Broader frameworks such as Multi-Criteria Decision Analysis,

reviewed by Cicciù et al. (2022) and Yuan et al. (2022), provide systematic ways to balance productivity, environmental sustainability, and economic viability. While these studies prove the potential of ML and decision-support systems, their application in the Philippines context—particularly Northern Mindanao—remains scarce, and most models have yet to be validated in diverse cropping systems beyond rice and maize.

The technological foundation for these innovations is strengthened by the integration of soil health sensors and IoT systems. Yin et al. (2021) emphasized the need for reliable, low-power, and scalable soil sensing technologies, while recent Philippine-based studies such as those by Sondhiya and Singh (2024) in Bukidnon and Kumar et al. (2024b) in rice and maize systems demonstrated the practicality of IoT-enabled NPK and pH sensors in monitoring nutrient status. Despite their promise, these sensor applications are still largely crop-specific and small-scale, limiting their ability to provide generalized, multi-crop recommendations across Region 10’s diverse agricultural systems.

The literature review establishes a strong foundation for developing a data-driven agricultural framework in Northern Mindanao. Soil health monitoring, zoning approaches, machine learning models, and IoT-based sensors converge to form comprehensive solutions. However, there remains a clear gap:

existing studies often treat these components separately—soil fertility studies without advanced analytics, zoning without integration to real-time data, or ML models without validation in local contexts. This study aims to address these gaps by developing an IoT-based Soil Health Monitoring and Recommendation System that integrates real-time soil parameters (NPK, pH, salinity, moisture, and temperature), agri-weather data, and agricultural zoning maps with machine learning. Unlike previous studies that focused on single crops or generalized recommendations, this system will deliver zone-specific and season-specific recommendations for both multi-cropping and fertilizer application in Region 10. By targeting key crops such as maize (corn), mungbean (mongo), soybean, squash (kalabasa), sweet potato (camote), cassava, eggplant, tomato, the project seeks to optimize yields, reduce resource waste, and promote sustainable agriculture practices aligned with the region's diverse farming systems.

CHAPTER 3

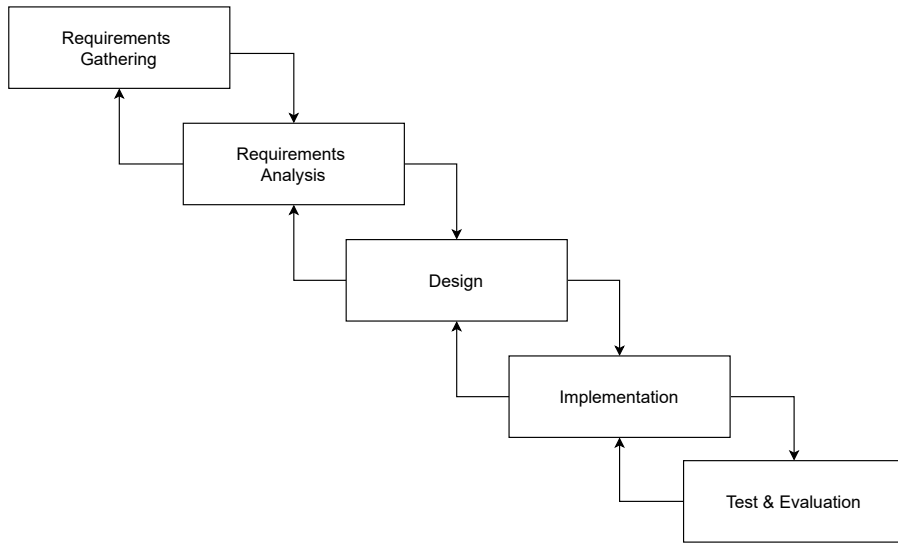
METHODOLOGY

In this chapter, the researchers outlined the methodology used to develop a soil health monitoring and recommendation system, which aimed to enhance sustainable agricultural practices through present data. The study utilized an experimental approach, integrating IoT sensors and machine learning algorithms, to enable continuous and accurate assessment of soil conditions. By focusing on essential soil parameters such as nitrogen, phosphorus, potassium, pH, salinity, moisture, and temperature this methodology provided insights that supported optimal crop management. This chapter also addressed the rationale behind the chosen experimental design, describing each stage of data collection, system setup, and analysis.

3.1 Research Design

Figure 3.1

Modified Waterfall Model



This study follows a modified waterfall research model to guide the development of the IoT-based crop and fertilizer recommendation system shown in Figure 4. The process begins with the requirements gathering phase, in which the system collects comprehensive data from the Department of Agriculture (DA) and credible online datasets. These datasets include zoning information, soil parameters, and crop-specific growth requirements. All collected datasets are integrated with the DA's zoning information to provide spatial context for agricultural analysis.

The study then proceeds to the requirements analysis phase, which identifies the need for soil health monitoring, crop prediction, and fertilizer recommendation in selected agricultural fields in Region~10. A gradient boosting algorithm is employed using both sensor-derived soil data and zoning information as inputs. Soil health prediction focuses on classifying soil quality levels, while crop suitability and yield forecasting utilize an optimized machine learning model to estimate expected productivity based on soil and environmental conditions. Fertilizer recommendations are generated based on predicted soil nutrient deficiencies and crop requirements derived from the machine learning analysis.

The next phase is system design, where the overall system workflow is defined, providing detailed descriptions of how the IoT-based machine learning integration operates. This phase presents the system architecture, hardware and software flowcharts, and the design of the web and mobile interfaces. The design ensures seamless integration of hardware components for data acquisition, software modules for data preprocessing and prediction, and user interfaces for monitoring and decision support.

This is followed by the implementation phase, which involves the development of the IoT system, training and testing of the machine learning model, and configuration of the required hardware and software components.

Field nodes collect and preprocess soil data before transmitting it to a central server for storage and predictive analytics. The system generates crop suitability and fertilizer recommendations, which are displayed through a web-based dashboard that visualizes soil health status, nutrient trends, weather information, farmer data, and zoning maps.

Once implementation is completed, the testing phase is conducted in the selected study area, where soil data are collected and prediction accuracy is evaluated. Zoning-based crop and fertilizer recommendations are validated against reference data and observed field conditions. The validation phase ensures system reliability through quantitative accuracy assessment and qualitative feedback on usability. Finally, the deployment and maintenance phase outlines potential large-scale application across Region~10, including recommendations for system improvement, scalability, and long-term sustainability.

This study follows a modified waterfall research model to guide the development of the IoT system shown in figure 4. The modified waterfall model allows an iteration to the previous phase for verification and validation, intended mainly for connecting stages. The process begins with the requirements gathering, in which the system will collect comprehensive data from the Department of Agriculture (DA) and credible online datasets. The datasets include zoning information and soil parameters, as well as crop-specific re-

quirements for growth conditions. Moreover, all datasets will be integrated with the DA's zoning information.

It then proceeds to the requirements analysis, which identifies the need for soil health monitoring, crop prediction, and fertilizer recommendation in the selected field in Region 10. A gradient boosting algorithm will be employed using both sensor and zoning data inputs. Soil health prediction will classify soil quality levels, while crop yield forecasting will leverage an optimized machine learning model with metaheuristic-based feature selection to predict expected productivity. Fertilizer recommendations will be generated based on predicted soil nutrient deficiencies, crop requirements, and established agronomic thresholds derived from sensor and zoning data. Feedback in this stage may return to the requirements gathering phase if analysis reveals missing data, incorrect assumptions, or the need for additional crop or soil parameters to properly define system requirements.

The next phase is the system design, where the flow of the overall system is defined with detailed descriptions on how the system will work on implementing the IoT-based ML Integration system. This phase involves a comprehensive overview of the system design, architecture, the flowchart of the hardware and software that will be used in this system, as well as the web and mobile interfaces. The design ensures seamless integration of hardware

components for data acquisition, software modules for data analysis and prediction, and user interfaces for monitoring control. The feedback may return to the requirements analysis stage if the proposed design cannot fully implement the defined requirements such as limitations in hardware integration or software module capabilities.

This is followed by the implementation phase, which involves the development of the IoT system, training and testing of the machine learning model, in combination with multi-criteria decision analysis methods, and the configuration of hardware and software. Field nodes gather and preprocess soil data, then transmit it to a central server for storage, predictive analytics, and generation of crop suitability and fertilizer recommendations. A web dashboard visualizes soil health, farmer data, nutrient trends, weather, and zoning maps. The results in this stage can return to the system design phase if there are software or hardware integration challenges encountered, and it requires adjustments in design to ensure proper functionality.

Once completed, the testing phase is carried out in the selected area where soil data is gathered and prediction accuracy is measured. In this phase, the zoning-based recommendations are also validated. The validation phase then ensures reliability through quantitative accuracy testing and qualitative feedback on usability. Feedback from testing may return to the implementation

phase to correct software bugs, or adjust hardware configurations for better performance before validation.

3.2 Requirements Gathering

The system requires comprehensive data collection to provide accurate, multi-cropping and fertilizer recommendation based on farming zoning in Budkinon, Philippines. Data will be sourced from the Department of Agriculture (DA), and credible online datasets. This includes detailed zoning information and soil parameters. To be included also are crop-specific requirements and growth conditions. Gathering these datasets ensures that the recommendation system is grounded in both environmental and agricultural best practices.

The types of data required encompass zoning information, and detailed crop profiles. Zoning data defines farming area, and land use classifications, while soil health parameters includes pH, NPK, salinity, moisture, and temperature. To be included as well is climate data such as historical rainfall, temperature, humidity, and forecast models which provide insights into environmental conditions that affect crop growth.

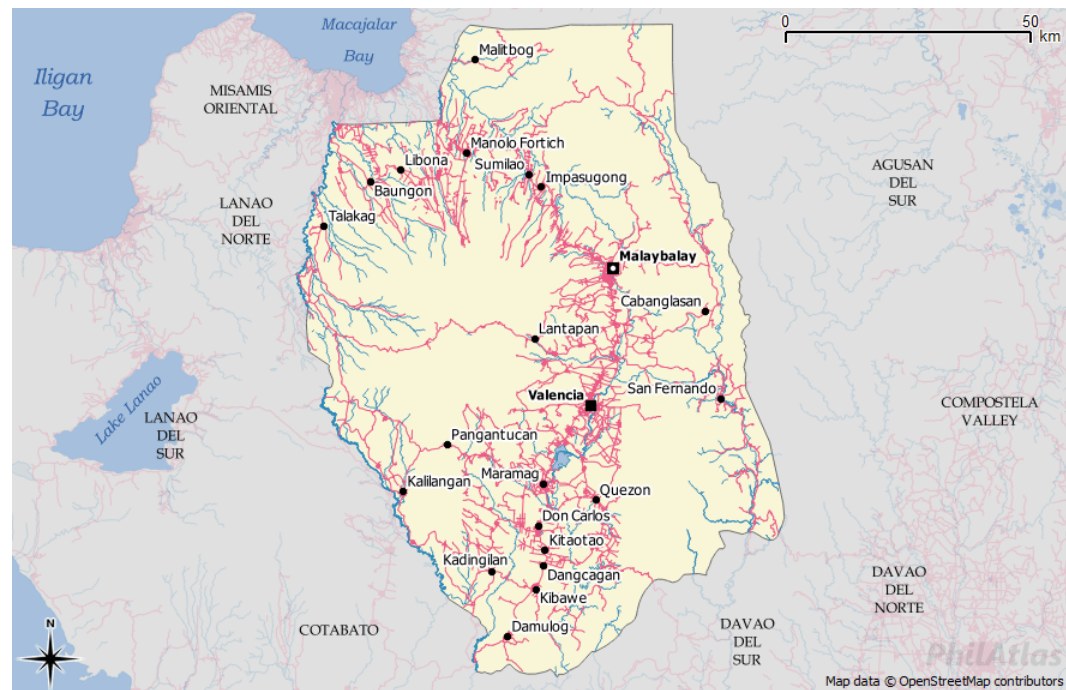
In preparation for system integration, the data will undergo a series of preprocessing steps. Missing sensor readings and data gaps will be addressed through interpolation, while noisy or incorrect data will be identified and re-

moved to maintain integrity. Soil and climate parameters will be normalized into uniform ranges to enhance the performance of predictive models. Once cleaned and standardized, all datasets will be integrated with the Department of Agriculture’s zoning information to ensure that recommendations are localized and relevant to the specific conditions in Bukidnon.

3.2.1 Research Setting

Figure 3.2

Research Setting (PhilAtlas)



The study will be conducted on specific farms in Region 10, specifically farms in Bukidnon, Philippines. The region is mostly agricultural, with vast lands focused on the cultivation of high-value crops such as maize (corn), mungbean (mongo), soybean, squash (kalabasa), sweet potato (camote), cassava, eggplant, tomato. The farms in this region typically employ conventional and semi-modern farming practices, which offers a realistic setting to monitor and assess crop performance under actual farming conditions.

3.3 Requirements Analysis

The system requires the development and implementation of a machine learning component capable of performing soil health prediction, and crop recommendation. Machine Learning boosting algorithms will be employed for these tasks, using both sensor and zoning data. The sensors will collect important soil information such as Nitrogen (N), Phosphorus (P), Potassium (K), pH level, moisture, temperature, and salinity. At the same time, zoning data from the Department of Agriculture will provide information about the type of land, elevation, and climate conditions of the area. By combining these two data sources, the system can better understand the relationship between soil condition and the environment. Based on the collected data, the machine learning model will classify soil quality levels, while crop yield forecasting will

leverage an optimized machine learning model by identifying patterns and relationships between soil parameters and environmental factors. To make predictions more accurate, the model will use a metaheuristic-based feature selection that focuses on the most relevant data and removes unnecessary or less useful information.

After analyzing the soil health parameters and producing crop recommendations, the system will generate fertilizer recommendations based on predicted soil nutrient deficiencies, crop nutrient requirements, and established agronomic guidelines. The system will determine the appropriate fertilizer type and application rate suited for specific field conditions. To make sure the machine learning model produces reliable results, the dataset will be divided into three parts: 70% for training, 15% for validation, and 15% for testing. This ensures that the system learns effectively from the data, maintains accuracy, and avoids overfitting.

The system will undergo proper evaluation using both quantitative and qualitative approaches. Quantitative assessment will include metrics such as accuracy, precision, recall, and F1-score for soil health prediction. R^2 , RMSE, and MAE for crop yield forecasting and consistency ratios and ranking stability indices for crop recommendation. Conducting these metric evaluations is important because they measure how well the machine learning model per-

forms and how reliable its predictions are. By analyzing these values, the researchers can identify the model's strengths, detect errors, and make adjustments to improve performance before deployment. Cross-validation will ensure model robustness across different crop zones. Meanwhile, qualitative evaluation will involve focus group discussions and structured interviews with farmers and agricultural technicians in Region 10 to assess usability, reliability, and practical value. Feedback from these sessions will guide further refinement of the system to ensure its effectiveness in real-world agricultural applications.

3.3.1 User Definition

Following the requirements analysis, the system's users were defined as follows:

Farmer - Farmers are the primary beneficiaries of the system. Their farms provide localized data inputs through the soil sensor. In return, farmers receive soil health insights, and multi-cropping recommendations.

Department of Agriculture (DA) - The DA serves as both a data provider and a supervisory user of the system. They supply zoning information, and other agricultural zoning datasets that form the foundation of the recommendation system. Through the web-based dashboard, DA officials can visualize regional soil health trends, and monitor farming zones.

The specific user requirements are summarized in Table 1.

Table 2: User Requirements

User	Requirements
Farmer	- Access soil health insights (pH, moisture, nutrients) through a mobile app. - Receive crop suitability scores and multi-cropping recommendations. - Get fertilizer schedules tailored to soil condition and crop type.
Department of Agriculture	- Provide zoning maps, soil datasets, and crop profiles to the system. - Access a centralized web dashboard to visualize soil health, nutrient trends, and farmer data across Region 10.

This table summarizes the requirements of the system’s two main users.

Farmers need accessible insights and recommendations to guide timely crop and fertilizer decisions, while the Department of Agriculture requires integrated data and analytics to support monitoring, reporting, and regional agricultural planning.

3.3.2 System Requirements

The system requirements shown in Table 2, were categorized into Input, Process, Output, Control, and Performance specifications to aid the system design and implementation.

Table 3: System Requirements

Category	System Requirement
Input Requirements	- The system shall accept input data directly from farmers and agricultural authorities through the designated interface. - The system shall acquire and process soil data from integrated sensors or external data sources. - The system shall accept soil health data from the sensor as input for hardware operation. - The system shall retrieve weather information from external APIs. - The system shall define sensor communication parameters to ensure consistent and accurate data acquisition from IoT nodes.
Process Requirements	- The system shall combine weather, soil, and farmer-provided data into a single decision-making framework. - The system shall employ boosting algorithms to enhance the accuracy of predictive crop recommendations. - The system shall analyze data-driven insights to balance yield potential with resource availability. - The system shall determine an appropriate planting timeline based on environmental conditions and crop growth requirements.
Output Requirements	- The system shall present crop recommendations via a visual interface on mobile and web applications. - The system shall display multi-cropping options, which shows compatible crop pairings. - The system shall present localized weather forecasts. - The system shall provide soil health reports with clear visual indicators. - The system shall display information on current crops cultivated by the farmer.

Category	System Requirement
	- The system shall display farm details of the farmer. - The system shall display the locations and overview of registered farms. - The system shall display a list of all farmers registered. - The system shall present crop rules, including crop rotation guidelines, and companion planting practices.
Control Requirements	- The system shall provide secure login and authentication for farmers and agricultural authorities. - The system shall restrict access to sensitive data based on user roles and permissions. - The system shall allow administrators to update crop rules and farm locations. - The system shall protect sensor and farmer data during transmission, and storage using standardized encryption methods and user authentication protocols.
Performance Requirements	- The system shall scale efficiently to accommodate growth in registered farmers and farms without requiring major redesign. - The system must be able to have offline syncing for farmer features. - The system must provide accurate crop recommendations for farmers. - The system shall synchronize data reliably between mobile and web platforms.

The system requirements were defined to ensure the effective collection, processing, and presentation of farm and soil data to support informed crop management decisions. The primary objective of the system is to assist farmers by integrating farm-specific information and soil health metrics

to generate accurate and actionable crop and fertilizer recommendations. Input requirements focus on collecting data from farmers and IoT-based sensors, while process requirements involve the use of advanced machine learning techniques, particularly gradient boosting algorithms, to analyze soil conditions, predict crop suitability, and estimate potential yield performance. Outputs are delivered through mobile and web applications to provide farmers and administrators with clear, visualized recommendations and reports. Control requirements ensure secure access through role-based permissions and administrative functions for managing crop data and farm records. Lastly, performance requirements address scalability, offline data synchronization, and recommendation accuracy to ensure reliable decision support across diverse agricultural environments.

3.4 Data Collection and Processing

This section describes the procedures and methods used to gather, integrate, and prepare data for the system's operation and machine learning components. It covers the collection of soil, environmental, and geographical data from various sources, as well as the acquisition of historical datasets. Furthermore, it outlines the processes involved in data integration, preprocessing, feature engineering, and dataset preparation to ensure data quality and suit-

ability for analysis. The section also presents how the processed data is stored and managed, and how it flows through the system to support accurate crop and fertilizer recommendations.

3.4.1 Soil Data Collection

Soil data collection serves as the primary input of the system, providing essential information for crop recommendation and soil health analysis. In this study, soil data were collected through a combination of IoT-based sensors and secondary data from the Department of Agriculture (DA) Region 10 to ensure both real-time and historical coverage.

The IoT sensing unit, integrated with an ESP32 microcontroller, measures key soil parameters such as nitrogen (N), phosphorus (P), potassium (K), soil pH, moisture, temperature, and salinity, which are critical in determining soil fertility and crop suitability. Data collection follows a periodic sampling approach, where readings are captured at set intervals and stored digitally for consistency and analysis.

To enhance data reliability and coverage, historical soil datasets and laboratory records from DA Region 10 were also incorporated. This combination of real-time sensor data and secondary sources ensures a more comprehensive and representative dataset for subsequent processing and machine

learning tasks.

Overall, the collected soil data form the foundation of the system, enabling accurate analysis, crop prediction, and fertilizer recommendation.

3.4.2 Environmental and Weather Data Collection

Soil and simple environmental readings from the field are sent to the ML service as one list of numbers. The first values match what the crop model was trained on (organic matter, phosphorus, potassium, and pH). If the device sends more values, those can include things like temperature, moisture, and electrical conductivity. After each prediction, the system saves a pending snapshot that holds the numbers, the full request, an optional farm id, and optional latitude and longitude. The app can then pull that single snapshot instead of combining data from many separate calls.

Weather works differently. The project does not use a weather station on the farm. The backend talks to OpenWeatherMap using lat and long, keeps the API key on the server, and sends the app a clean package with temperature, humidity, pressure, wind, visibility, a short description of the sky, the place name, and a time stamp. The app does not need to hit the weather API constantly, so it can cache results for a short time. Which coordinates are used matters: if the latest ML request already had lat and long, those are

used for weather so it matches the same area as the sensor run. If not, the app uses the coordinates stored for that farm. There is also an optional fallback for testing when coordinates are missing. When the farmer has an active saved session, the screen shows the saved soil data from the server for consistency. When there is no active session, the app checks often for new pending readings so the numbers stay up to date.

This setup is intentional. Soil measurements and internet weather are not the same thing, but showing both together gives a clearer picture. Putting the weather API on the backend hides the key and keeps responses in one standard format. Using one feature list and one pending snapshot keeps the model, the server, and the app on the same page and avoids extra work in the mobile client.

3.4.3 Zoning and Geographical Data Collection

Geographic representation matters so farm records are not just names on a list but real places on the ground. The system needs to know where farming happens, roughly how big the holding is, and stable coordinates that other tools can use, such as weather APIs and machine learning workflows. Right now, location is stored mostly as a point: each farm can have latitude and longitude, plus an optional text field for things like barangay or street,

and a field for how large the farm is said to be. Admins and users enter this through the web when they create or edit a farm. The API can also update coordinates when a session or a sync from the model layer brings in new positions, so people do not have to fix the map by hand as often.

In this study, the term “zoning” means organizing farms in space, not official land-use zones drawn as polygons on a cadastral map. The admin GIS Mapping screen shows farms on a Leaflet map with OpenStreetMap tiles. Farms with real latitude and longitude show up as pins. Farms without coordinates stay off the map until someone adds GPS data, and the UI warns you if registered farms still have no usable GPS. If there is more than one farm, the map zooms to fit all the pins. The default map center fits the Philippines so the first view is not empty. Clicking a pin shows extra details like soil type and area so you can relate the site to soil conditions. Soil tests are linked to the same farm id, so lab or field results line up with that place in the dashboard

3.4.4 Historical and External Data Acquisition

The main purpose of historical data collection is to build a strong foundation for training the machine learning models that control the crop and fertilizer recommendations of the system. While the Internet of Things (IoT) sensor network, powered by the ESP32 microcontroller, is highly capable of

capturing local real-time data, machine learning algorithms need substantial historical context to recognize complex agricultural patterns. Therefore, additional historical data is collected directly from the primary dataset titled *Consolidated Soil Info (RSL) 2023 to 2025 USTP CPE4*.

This core document combines official soil analysis reports and agricultural data gathered by the Regional Soils Laboratory (RSL) of the Philippine Department of Agriculture spanning the years 2023 to 2025. The collected dataset contains vital features, specifically basic soil parameters such as Nitrogen, Phosphorus, Potassium levels, pH, moisture, and temperature. It also includes the corresponding crop types traditionally grown under those exact conditions and standard fertilizer baselines. The exact features extracted for the system database and their respective categories are detailed in **Table 4**.

Table 4: Tabular Summary of Features Extracted from the RSL Dataset

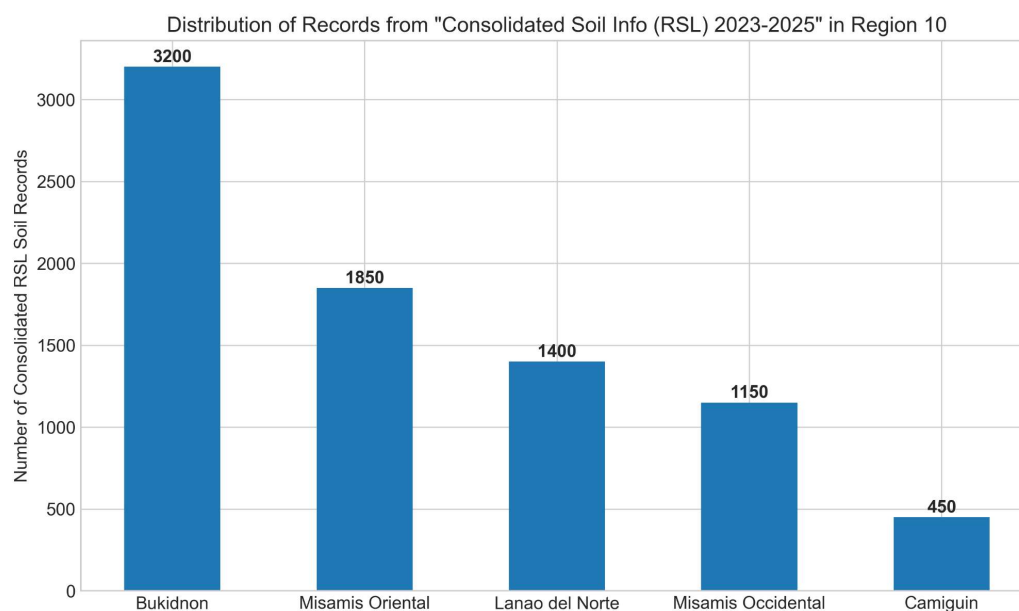
Category	Extracted Features
Soil Nutrients	Nitrogen (N), Phosphorus (P), Potassium (K)
Soil Physics	pH Levels, Moisture Content, Temperature
Agricultural	Crop Type, Fertilizer Baselines
Metadata	Date of Analysis, Provincial Location

To ensure the models can generalize well across local environmental conditions, the combined document includes a large number of distinct records to create a realistically sized data collection. Furthermore, strict data relevance

rules built into this dataset guarantee that the information is restricted exclusively to Region 10 (Northern Mindanao). This focuses specifically on its provinces, including Bukidnon, Misamis Oriental, Lanao del Norte, Misamis Occidental, and Camiguin. As illustrated in **Figure 3.3**, the distribution of these records across the five provinces reflects the agricultural density of the region, with Bukidnon providing the most significant volume of data.

Figure 3.3

Distribution of Soil Records across Region 10 Provinces



Before the data is formally entered into the primary database of the system, early validation procedures are performed. These involve automated limit checking against the laboratory records to confirm structural correctness

and data credibility. This process ensures that the dataset remains a high-quality reference for training the system’s recommendation engine.

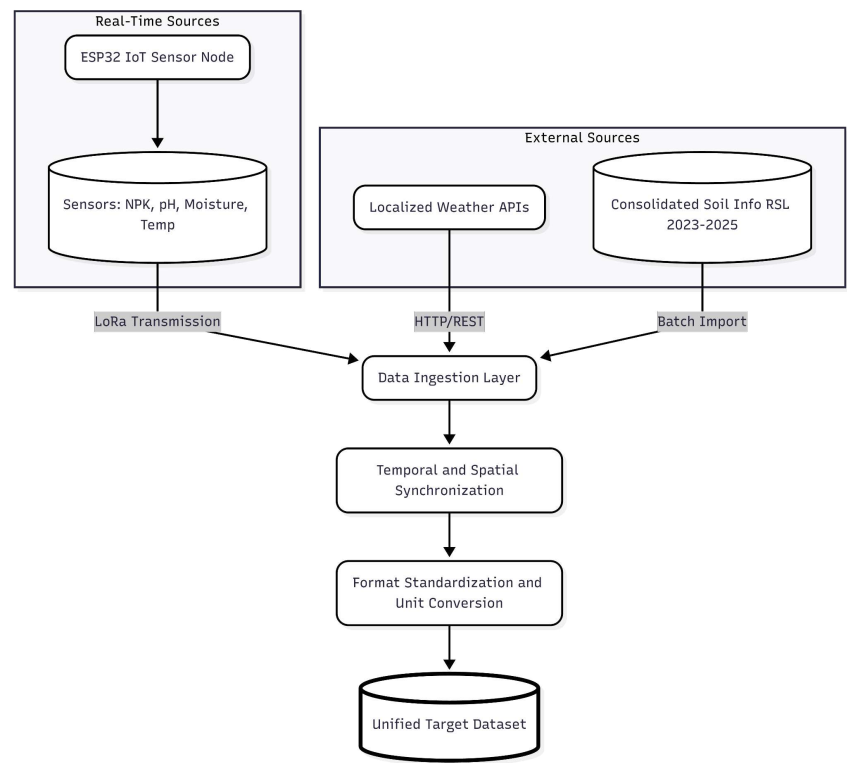
3.4.5 Data Integration

The primary objective of the data integration phase is to unify multiple, heterogeneous data streams into a single, cohesive dataset suitable for advanced computational analysis and machine learning training. By centralizing disparate information into a uniform structure, the system ensures that the predictive models can process environmental, historical, and real-time variables simultaneously. This integration is essential for creating a holistic representation of the agricultural environment, bridging the gap between local sensor observations and broader regional contexts.

As illustrated in the integration pipeline in **Figure 3.4**, the system aggregates data from both real-time and external sources. Real-time data is captured via an ESP32 IoT sensor node, which transmits continuous readings of Nitrogen (N), Phosphorus (P), Potassium (K), pH levels, soil moisture, and temperature. These hardware-derived metrics are combined with external data streams, including localized weather updates retrieved through HTTP/REST APIs and the validated Consolidated Soil Info (RSL) 2023–2025 historical dataset. This multi-source approach allows the framework to factor in both

immediate soil conditions and long-term agricultural trends.

Figure 3.4
System Architecture of the Data Integration Pipeline



Note. The pipeline depicts the convergence of LoRa-based sensor data, REST-based weather APIs, and batch-imported historical records into a unified dataset.

To ensure contextual accuracy, the system employs rigorous temporal and spatial synchronization. Data ingestion occurs through different protocols depending on the source: LoRa transmission for hardware sensors, REST-ful requests for weather data, and batch imports for historical records. Once ingested, these data points are programmatically aligned based on match-

ing timestamps and geographic coordinates. This synchronization forms a complete environmental snapshot, ensuring that a specific sensor reading is correctly paired with the weather conditions and historical soil health data relevant to that exact time and location.

During the aggregation process, structural inconsistencies and varying data formats are resolved through a dedicated format standardization and unit conversion layer. Because different sources often utilize varying measurement scales, the system applies strict conversion rules to ensure uniformity. For example, temperature readings are standardized to Celsius, and moisture levels are normalized to percentage values. Additionally, schema mapping is performed to align field names across the various data inputs, ensuring that all variables conform to a consistent naming convention.

The culmination of this pipeline is the creation of a Unified Target Dataset. This centralized repository effectively bridges real-time sensor baselines with extensive historical context and atmospheric data. By outputting a standardized, synchronized matrix of information, the integration phase provides the high-fidelity data foundation necessary for the subsequent preprocessing and machine learning stages of the system.

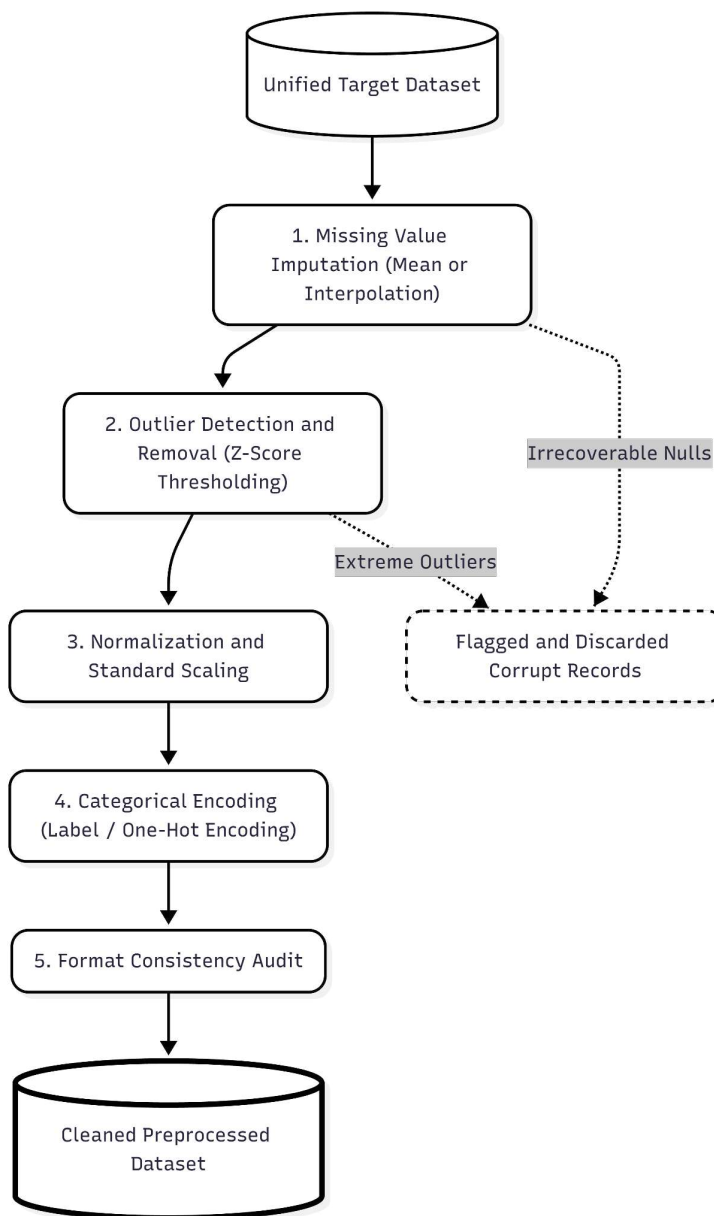
3.4.6 Data Preprocessing

The primary objective of data preprocessing within this framework is to systematically refine and clean the integrated dataset, thereby reducing algorithmic bias and ensuring high accuracy during the subsequent machine learning training phase. Given that the machine learning models rely on the structural integrity of the input data to generate reliable crop and fertilizer recommendations, this phase acts as a critical filter. It transforms raw, heterogeneous data into a standardized format that facilitates efficient mathematical computation and pattern recognition.

The initial stage involves rigorous data cleaning to address missing or empty values. As illustrated in the sequential pipeline in **Figure 3.5**, these gaps are handled based on the statistical variation of the feature. Depending on the feature's distribution, these gaps are addressed through mathematical substitution, such as applying the feature mean or using linear interpolation, or through the systematic removal of records showing excessive missing data to maintain dataset integrity.

Figure 3.5

Sequential Data Preprocessing Pipeline for Agricultural Soil Records



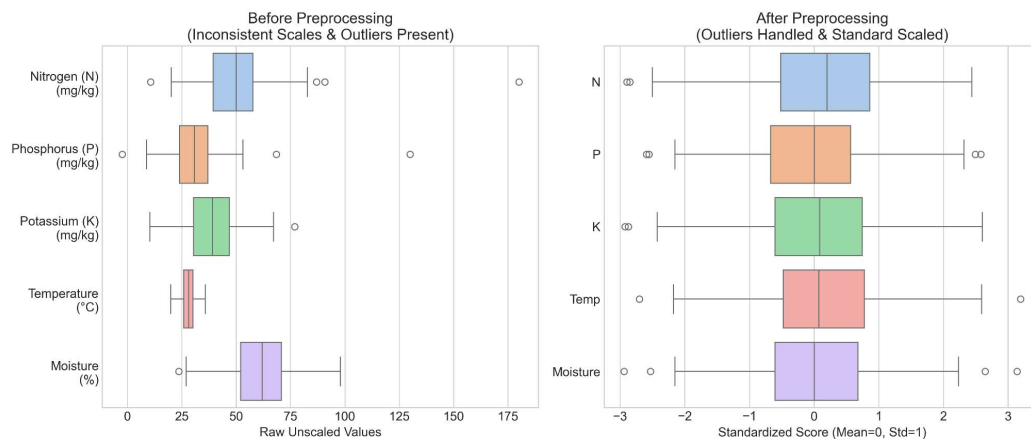
Following the initial cleaning, noise and outlier handling mechanisms are applied to filter unusual data points generated by temporary sensor glitches

or transmission errors. This process employs statistical limits, specifically Z-score thresholding, to identify and discard extreme values that could otherwise skew the decision boundaries of the model. The impact of this process is evident in Figure 3, where the "Before Preprocessing" box plots indicate several extreme outliers in the Nitrogen (N) and Phosphorus (P) features. By removing these points, the system ensures that the training process is grounded in realistic agricultural conditions rather than technical anomalies.

To ensure that variables with naturally wide numerical ranges do not disproportionately dominate the learning algorithm, normalization and scaling techniques are utilized. As shown in the comparison in **Figure 3.6**, soil parameters such as Potassium (K) and Nitrogen (N) often exist on scales significantly different from Temperature or Moisture. Using a standard scaler, these continuous numerical features are transformed into a common scale where the mean is 0 and the standard deviation is 1. This standardization is crucial for the stability and convergence of the machine learning algorithms, as it allows the model to assign appropriate weights to each feature without being influenced by its raw magnitude.

Figure 3.6

Comparison of Soil Parameter Distributions Before and After Preprocessing



Algorithmic encoding is performed to translate categorical variables into numerical representations. Since the system’s backend requires mathematical inputs for computation, non-numeric data—most notably the targeted crop types—are transformed using techniques such as label encoding or one-hot encoding. This process ensures that categorical labels are represented as distinct mathematical entities, allowing the model to process variety-specific data without assuming an incorrect ordinal relationship between different crops.

The preprocessing pipeline concludes with comprehensive programmatic consistency checks to definitively verify the structural integrity and uniformity of the cleaned dataset. This final audit ensures that all processed records adhere to the predefined schema and that no data corruption occurred during the transformation stages. Once verified, the dataset is formally transitioned

into the primary database, serving as a high-fidelity reference for the system's recommendation engine.

3.4.7 Feature Engineering

The primary objective of the feature engineering phase was to augment the predictive performance of the machine learning models by providing more nuanced and expressive data representations. Rather than relying exclusively on the raw soil properties captured through the sensor network and historical datasets, this phase involved mathematically deriving new variables that encode established agronomic relationships. By transforming raw measurements into high-order features, the system enables the algorithms to capture complex, non-linear dependencies between soil chemistry and plant health that are not explicitly present in the initial input vectors.

Two distinct categories of derived features were constructed for the engineered dataset. The first category comprised ratio-based features designed to quantify the balance between specific nutrients. The Phosphorus-to-Potassium ratio (P/K) was calculated to define the internal nutrient equilibrium between these two primary macronutrients, which serves as a critical fertility indicator in tropical soil profiles. Additionally, the Phosphorus-to-Organic Matter ratio (P/OM) was computed to express Phosphorus availability relative to the

soil’s organic composition, reflecting the established agronomic dependency of Phosphorus mobility on organic matter levels.

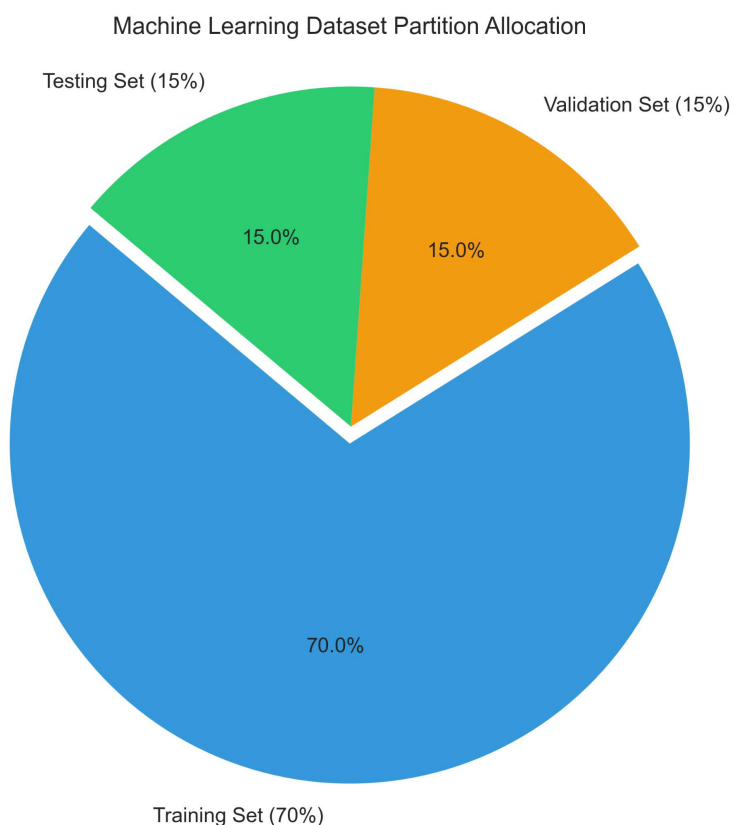
The second category consisted of interaction features, derived through multiplicative operations to encode conditional relationships. An Organic Matter $\times$ pH interaction term was generated to represent the principle that organic matter decomposition and subsequent nutrient release are strongly regulated by soil acidity. Similarly, a Potassium $\times$ pH interaction term was derived to capture the pH-dependent nature of Potassium uptake by root systems. These interaction terms allowed the model to learn complex interdependencies without requiring manual thresholding or excessively deep neural architectures.

The explicit encoding of these soil-crop relationships reduced the computational complexity required for the model to discover such patterns independently. By providing calculated nutrient ratios and pH-dependent interaction terms, the algorithm leveraged domain-specific knowledge directly, resulting in faster convergence during the training phase and improved generalization when processing unseen soil conditions. This engineering process successfully expanded the dataset from the four original raw variables to a comprehensive set of eight predictive features.

3.4.8 Dataset Preparation for Machine Learning

The primary objective of dataset preparation is to establish the final structural configuration of the preprocessed data, ensuring it is partitioned and formatted for the iterative training and evaluation cycles of the machine learning models. This phase represents the transition from static data storage to an active learning format, where information is structured to minimize bias and maximize the model's ability to generalize across diverse agricultural scenarios.

A fundamental component of this phase is dataset splitting, where the unified data is stratified to maintain representative class distributions across all subsets. As illustrated in **Figure ??**, the dataset is divided into three definitive partitions: 70% of the data is allocated for direct model training, 15% is utilized for validation to systematically tune hyperparameters, and the remaining 15% is strictly reserved for blind testing to evaluate final real-world performance.

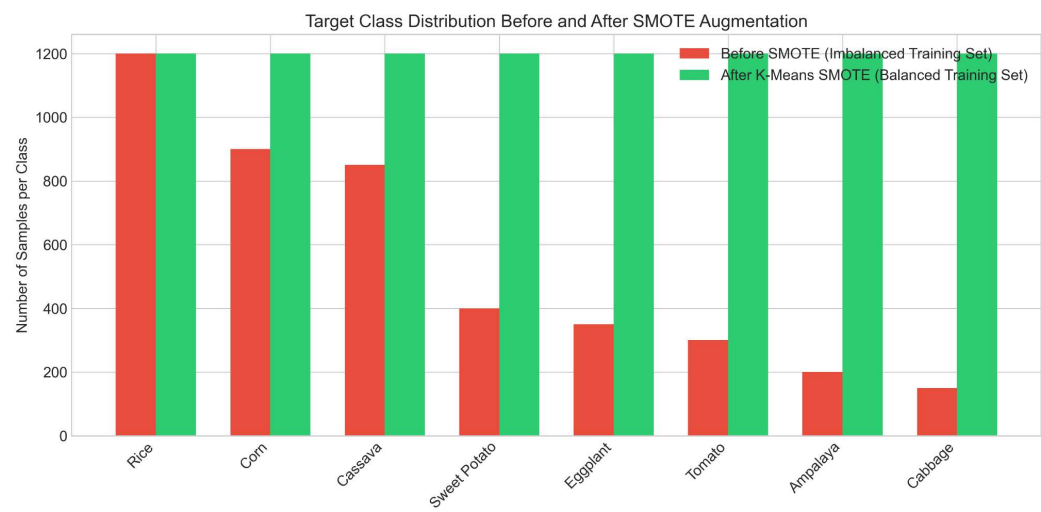
Figure 3.7*Machine Learning Dataset Partition Allocation*

Note. The 70-15-15 split ensures sufficient data for training while maintaining independent sets for tuning and final evaluation.

Concurrent with this partitioning is the execution of precise data labeling. The feature space, which includes the eight engineered soil and environmental metrics, is distinctly separated from the target variables. This frames the system's objectives into two distinct paths: crop recommendation is treated as a discrete multiclass classification problem, while fertilizer recommendations are mapped as specific quantitative output variables.

To address potential disparities in crop representation within the historical data, advanced class imbalance handling is applied. The system utilizes the K-Means Synthetic Minority Oversampling Technique (SMOTE) on the training subset to avoid predictive bias toward majority classes. The effect of this augmentation is clearly demonstrated in **Figure 3.8**, where the imbalanced classes are synthetically oversampled to reach a uniform baseline of 1,200 samples per class, ensuring the model learns the characteristics of all crops with equal weight.

Figure 3.8
Target Class Distribution Before and After SMOTE Augmentation



Note. SMOTE application resolves class imbalances by generating synthetic samples for minority classes like Cabbage and Ampalaya.

3.4.9 Data Storage and Management

The architecture of the proposed system necessitates a centralized and reliable repository to ensure data consistency across administrative and end-user platforms. This repository facilitates the management of diverse entities, including farmer and administrator accounts, farm records encompassing geographic coordinates and land area, active farming sessions, and soil test readings. Furthermore, the database stores the outputs of the machine learning engine, specifically crop and fertilizer recommendations, alongside system telemetry and device status updates. While machine learning models are maintained as distinct training artifacts, all operational and longitudinal data are housed within the database to ensure that the web dashboard, mobile application, and backend services utilize a unified source of truth.

The system utilizes a standard relational database management system, specifically PostgreSQL, which is deployed via a managed cloud service to ensure high availability and automated maintenance. The backend Application Programming Interface (API) establishes a secure connection to the database, organizing data into structured tables for users, farms, sessions, soil tests, and recommendations. To optimize schema complexity, richer nested data structures are stored using structured fields within the relational environment, reducing the need for an excessive number of auxiliary tables while

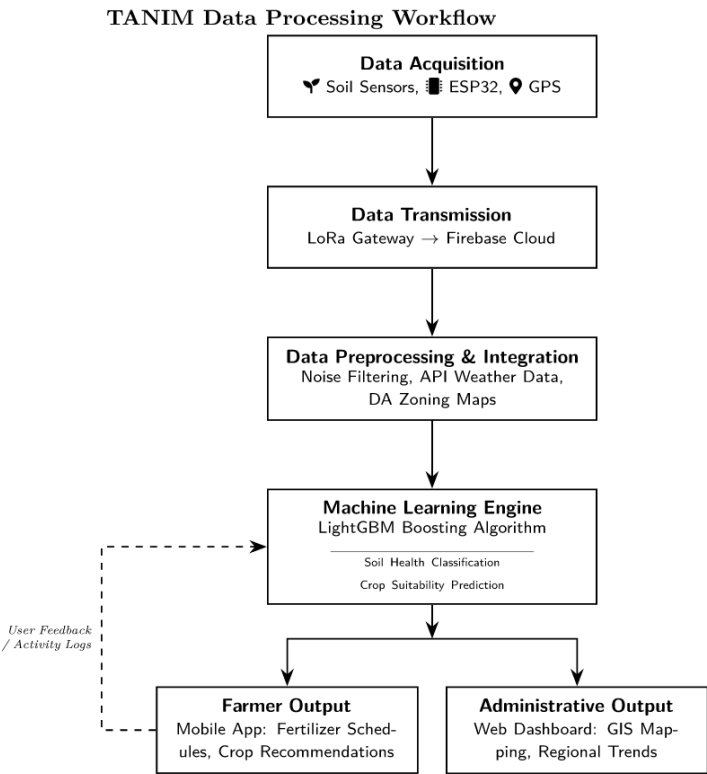
maintaining query efficiency. The mobile application interacts with this layer through standard HTTP requests, while the web application leverages a Representational State Transfer (REST) layer provided by the hosting service to streamline Create, Read, Update, and Delete (CRUD) operations without schema duplication.

Security and scalability are integral to the data management strategy of the system. User authentication is secured through strong cryptographic hashing for passwords, while session management is handled via a dual-token system consisting of short-lived access tokens and longer-lived refresh tokens. These tokens are stored within secure on-device storage on the mobile platform and verified against secrets maintained in the server configuration. Furthermore, the hosting environment implements Row-Level Security (RLS) policies to ensure that users are restricted to accessing only their authorized data subsets. To accommodate increasing traffic and data volume, the system employs connection pooling, stateless API instances capable of horizontal scaling behind a load balancer, and paginated data retrieval to minimize latency and server overhead. Consistent data integrity is maintained through automated backup protocols managed by the database provider, ensuring recovery in the event of system failure.

3.4.10 Data Processing Workflow

The TANIM system employs a multi-stage data pipeline designed to transform raw environmental signals into actionable agricultural intelligence. This workflow integrates physical sensing, cloud-based data management, and advanced machine learning analytics to provide decision-support tools for both farmers and regional agricultural authorities. As illustrated in **Figure 3.9**, the comprehensive data flow, from initial acquisition to end-user dissemination, follows a logical progression through hardware interfacing, cloud integration, and predictive modeling.

Figure 3.9
TANIM Data Processing Workflow Pipeline



Note. The workflow demonstrates the transition from raw sensor acquisition to the generation of agricultural insights.

The primary stage of the workflow involves the high-fidelity collection of soil parameters at the physical layer. The system utilizes a multi-parameter soil sensor to monitor Nitrogen (N), Phosphorus (P), Potassium (K), pH, moisture, salinity, and temperature, all of which are interfaced with an ESP32 microcontroller. To ensure data integrity during local transmission, a MAX485 module is employed to convert RS-485 differential signals to TTL-compatible

logic for the microcontroller. These data packets are subsequently encapsulated and transmitted via the LoRa (Long Range) protocol to a central gateway. This modulation technique is specifically selected to ensure reliable connectivity in topographically challenging or remote agricultural zones where traditional cellular infrastructure is often insufficient.

Upon receipt at the gateway, the data is forwarded to a Firebase cloud database for centralized management and refinement. During this phase, the system performs rigorous data cleaning by executing filtering algorithms to mitigate electrical noise and remove statistical outliers from the raw sensor readings. Furthermore, a data fusion process contextualizes the real-time sensor data by integrating it with secondary datasets. These include meteorological information, such as precipitation and humidity acquired via external APIs, and geospatial zoning maps provided by the Department of Agriculture (DA) Region 10.

The analytical core of the system leverages a LightGBM algorithm, a gradient boosting framework utilized for its high efficiency and superior performance with tabular agricultural datasets. The process begins with metaheuristic-based feature selection to identify the most significant soil and climatic variables, thereby reducing dimensionality and improving model generalization. The predictive engine then performs dual-task analytics, which

include classifying soil health status based on established agronomic thresholds and forecasting crop suitability to optimize regional yield performance.

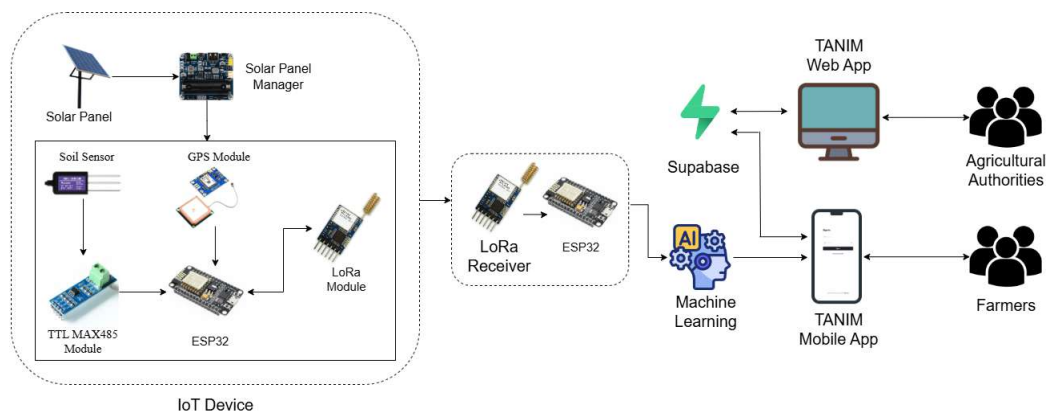
3.5 System Design

In this section, the flow of the overall system is defined with detailed descriptions on how the system will work on implementing the IoT-based ML integration system. This involves a comprehensive overview of the system design and architecture, and the flowchart of the hardware and software that will be used in this system. The design ensures seamless integration of hardware components for data acquisition, software modules for data analysis and prediction, and user interfaces for monitoring and control.

3.5.1 System Architecture

Figure 3.10

System Architecture



The figure 3.10 shows the updated system architecture for the Soil Health Monitoring, Fertilizer, and Crop Recommendation System (TANIM). The system goes from hardware in the field, through a LoRa link and a local gateway, into the cloud for storage and machine learning, and then out to a web app for agricultural authorities and a mobile app for farmers. Field IoT device. The sensing unit is basically one IoT package in the field. A solar panel feeds a solar panel manager, which handles charging and power so the device can keep running in areas that are far from the grid. Inside the box, an ESP32 reads the sensors, does any simple processing on the readings, and builds the packets that will be sent out.

A soil sensor measures things like NPK, pH, and moisture. The sensor talks over an industrial serial link that behaves like RS-485, which is good when the cable is long and the environment is noisy. A TTL MAX485 module turns that into a regular UART signal the ESP32 can read, so you get reliable communication without making the firmware too complicated. The device also has a GPS module. The ESP32 reads the GPS data and sends location together with the soil readings. That way the cloud and the TANIM Web App can show maps and trends by area, not just a list of numbers with no place attached.

The ESP32 sends data through a LoRa module. LoRa is meant for low

power and longer range, which fits farms and rural sites. Gateway and path into the cloud. A LoRa receiver listens for packets from one or many nodes in the field. It is connected to a second ESP32 that works as a gateway. That ESP32 receives the LoRa traffic, checks the data, can hold or reshape packets if needed, and then sends everything upstream. The forwarded data includes soil readings, and GPS location.

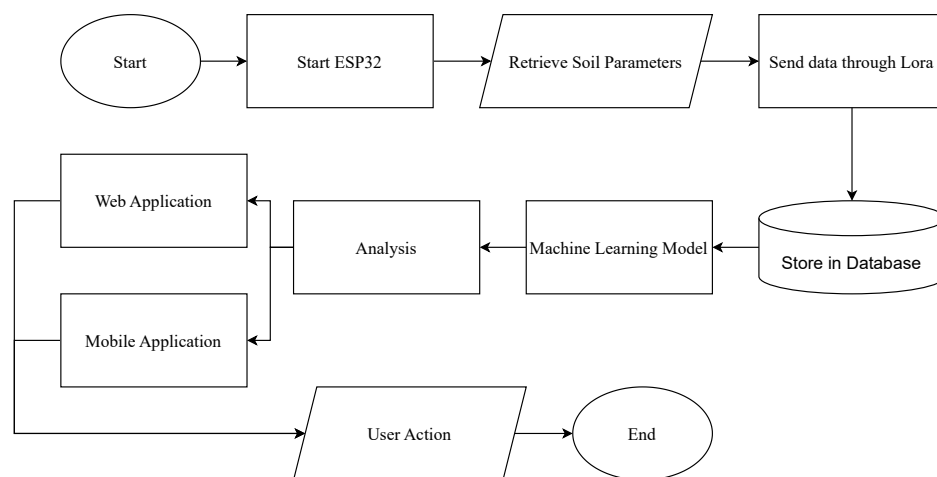
Cloud, machine learning, and apps. After the gateway, the system splits into two main parts in the cloud. One part is machine learning, which uses new data and past data to look for patterns in the soil, spot possible problems like low nutrients or moisture stress, and suggest fertilizers, crops, and simple soil health takeaways. The other part is Supabase, which is the main backend for the project. Supabase stores data in a database, handles sign in and users, and exposes APIs for things like sensor history, profiles, and who is allowed to see what, especially for authority users.

The TANIM Web App is built for agricultural authorities. It uses Supabase for the underlying data and adds views like dashboards and maps, plus results from the ML side where those are wired in. The TANIM Mobile App is built for farmers in plainer language: soil insights, recommendations, alert. Both apps use Supabase for secure access to stored data and connect to the machine learning side when the app needs model based recommendations.

3.5.2 Flowchart

Figure 3.11

System Flowchart



The figure 3.11 presents the operational workflow of the proposed soil health monitoring and recommendation system, highlighting the data flow from the field device to the end-user applications. The process initiates at the sensing stage, where the ESP32 microcontroller, powered through a solar energy management unit, interfaces with soil sensors to capture essential soil parameters, including Nitrogen, Phosphorus, and Potassium (NPK), pH, moisture, temperature, and soil salinity. These parameters are critical indicators of soil fertility and crop suitability. Once collected, the data is processed locally

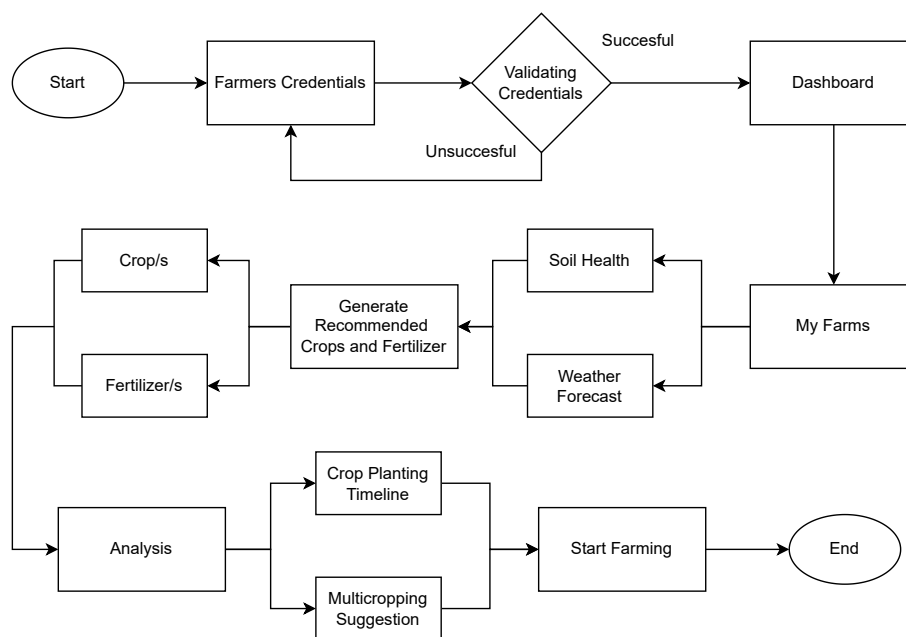
by the ESP32 to ensure efficient packaging before transmission. Leveraging LoRa technology, the device transmits the data wirelessly over long distances to a gateway, which serves as the intermediary link between the sensing unit and the cloud infrastructure. The gateway then forwards the collected data to a cloud database, ensuring reliable and scalable data storage for subsequent analysis.

Within the cloud environment, the raw soil data undergoes processing and analysis through machine learning models. These models are trained on historical soil and crop datasets to identify patterns, classify soil conditions, and predict nutrient deficiencies. By transforming raw sensor readings into interpretable outputs, the system generates actionable insights regarding soil health and crop management. The analytical results include fertilizer recommendations and crop suitability assessments, enabling evidence-based decision-making for end users. This layer of intelligence ensures that data collected from the field is not merely stored but is also transformed into knowledge that directly supports precision agriculture practices.

The processed insights are subsequently disseminated through two primary platforms: a web application and a mobile application. The web application is designed for agricultural authorities and extension officers, providing comprehensive monitoring dashboards, spatial analysis tools, and regional soil

health reports. This enables decision-makers to track soil conditions at scale, identify priority intervention areas, and develop region-specific agricultural strategies. The mobile application targets farmers as the primary stakeholders, offering a user-friendly interface to access real-time soil health status, recommended fertilizer dosages, and crop-specific advisories. Notifications and alerts are integrated into the system to provide timely guidance, particularly in scenarios requiring immediate action.

An integral component of the workflow is the feedback mechanism incorporated within both platforms. Farmers and users are able to submit observations, report crop outcomes, and validate system recommendations. This feedback not only enhances farmer engagement but also provides valuable data for iterative improvements of the machine learning models. As a result, the system establishes a continuous learning cycle wherein recommendations are refined over time, leading to greater accuracy and contextual relevance. Overall, the workflow described in Figure 3.11 demonstrates how the integration of IoT sensing, cloud computing, and machine learning facilitates a comprehensive, data-driven approach to soil health management and crop productivity enhancement.

Figure 3.12*Software Mobile Application*

The figure 3.12 illustrates the workflow of the farming recommendation system, highlighting the interaction between farmers and the application in obtaining crop suggestions tailored to soil and weather conditions. The process begins with user authentication, wherein the farmer logs into the system using their credentials. The application validates these credentials against the stored records in the database, ensuring secure access. If the credentials are incorrect, the system prompts the user to re-enter valid information, thereby safeguarding the platform against unauthorized access. Upon successful login,

the farmer is redirected to the dashboard, which serves as the central interface for accessing personalized farm data and system functionalities.

From the dashboard, the farmer navigates to the “My Farms” section, where the application consolidates and displays farm-specific data. This includes soil health parameters gathered from IoT-enabled sensors and environmental information derived from integrated weather forecasting services. The system then processes this combined dataset using its analytical and decision-making algorithms to generate a list of recommended crops that are most suitable for the prevailing soil and weather conditions. By leveraging both real-time and contextual data, the system ensures that the recommendations are accurate, adaptive, and relevant to the farmer’s specific location and situation.

In areas where internet connectivity is limited, the application supports an offline mode that allows farmers to continue accessing previously stored data and recommendations. This feature ensures uninterrupted usability by caching essential farm information, soil records, and the most recent crop recommendations on the device. Once the connection is restored, the system automatically synchronizes any new data and updates the stored records in the cloud. Through this mechanism, farmers in remote areas can still make informed decisions and manage their farms effectively, even without continuous

internet access.

Once the recommendations are presented, the farmer can review the options and select the crop that aligns with their goals, preferences, and available resources. Following this selection, the system provides advanced features designed to further support farm management. These include multicrop suggestions, which propose crop diversification strategies for risk mitigation and sustainable land use, as well as a detailed crop planting timeline that outlines optimal planting and cultivation schedules. Such features are aimed at enhancing farming efficiency, improving resource allocation, and maximizing yield potential.

The final stage of the workflow involves the farmer applying the recommendations provided by the system. By combining data-driven crop selection with structured guidance on crop management, the system enables farmers to initiate the farming process with greater confidence and scientific support. This workflow underscores the system's capacity to guide users from initial login and farm data access to informed decision-making and implementation in the field. Ultimately, the integration of secure user access, real-time soil and weather data, intelligent recommendation algorithms, and decision support tools demonstrates how the farming recommendation system contributes to improved agricultural productivity and long-term sustainability.

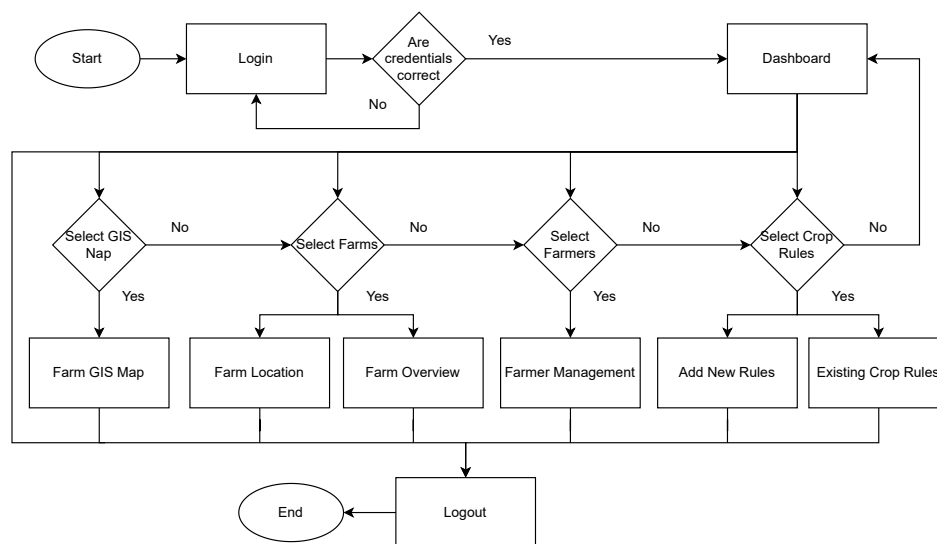
Figure 3.13*Web Application Flowchart*

Figure 3.13 illustrates the workflow of the Farm Management Web Application, which is designed to support agricultural administrators in managing and monitoring farming activities at a regional scale. The process begins with a secure login mechanism, where the administrator is required to enter valid credentials. The system verifies these credentials against stored records to ensure authorized access. If the credentials are invalid, the administrator is redirected back to the login page to re-enter the correct information. Upon successful authentication, the administrator is directed to the dashboard, which serves as the central navigation interface of the system.

From the dashboard, the administrator can access four primary modules represented by the decision nodes in the flowchart. The GIS Map module provides geospatial visualization of registered farms, enabling spatial analysis and regional monitoring. The Farms module supports the management of farm-related information, including farm locations and farm overviews, ensuring accurate documentation of farm records. The Farmers module focuses on farmer management, allowing administrators to organize farmer profiles and maintain farmer-related data. The Crop Rules module facilitates the administration of agricultural policies by enabling the addition of new crop rules and the review of existing crop regulations.

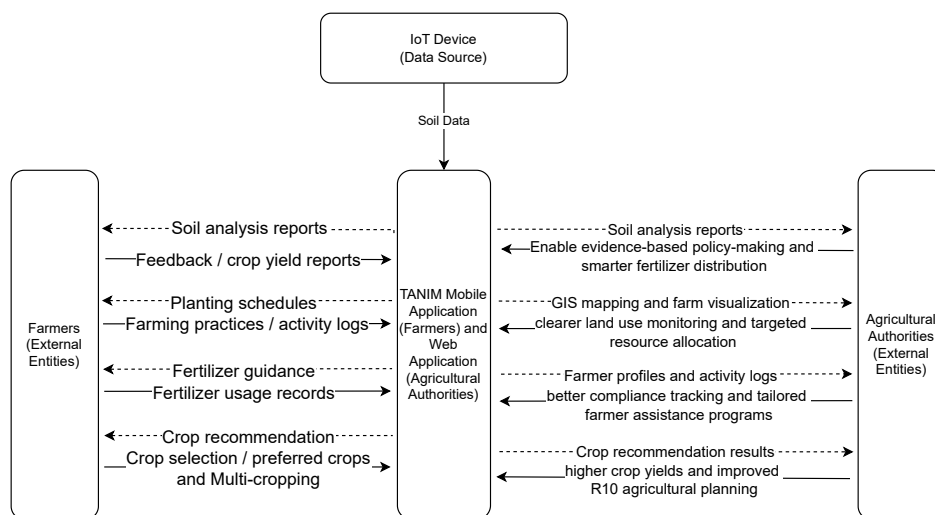
Each module is seamlessly integrated with the dashboard, allowing administrators to navigate efficiently between system functions. After completing tasks within a selected module, the administrator may return to the dashboard or securely terminate the session through the logout function. This workflow highlights the system's structured approach to access control, centralized administration, and policy management, supporting effective monitoring, data-driven decision-making, and sustainable agricultural governance for the Department of Agriculture.

3.5.3 Context Level Diagram

The Context-Level Diagram provides a high-level view of how the TANIM system interacts with its primary stakeholders: Agricultural Authorities and Farmers. It illustrates how data flows into and out of the system without detailing internal processes.

Figure 3.14

Context-Level Diagram Between the System And The External Entities



The Context-Level Diagram illustrates the integrated interaction of the TANIM system with both Agricultural Authorities and Farmers through its unified web and mobile applications. Agricultural Authorities do not directly generate raw data, instead, they receive processed insights derived from IoT

devices deployed in the field, including soil analysis reports, GIS-based farm visualizations, farmer profiles, and crop recommendation results. These outputs support evidence-based policymaking, targeted resource allocation, compliance monitoring, and improved agricultural planning. At the same time, IoT devices serve as primary data sources by providing soil and environmental information to the system, while farmers contribute essential inputs such as crop yield feedback, farming activity logs, fertilizer usage records, and crop preferences. TANIM processes this continuous exchange of data to produce actionable outputs such as planting schedules, fertilizer guidance, crop recommendations, and farming practice records. Through this unified data flow, the system transforms raw field data into actionable intelligence that supports sustainable farming practices, enhances farmer productivity, and enables informed, timely decision-making that contributes to national food security goals.

3.5.4 Data Flow Diagram

The Data Flow Diagram visually represents how data flows within the TANIM system. It highlights the interactions between IoT devices, farmers/agricultural authorities, and the system's core processes, demonstrating how data is retrieved, stored, and analyzed to provide soil health insights and

crop/fertilizer recommendations.

Figure 3.15

Data Flow Diagram

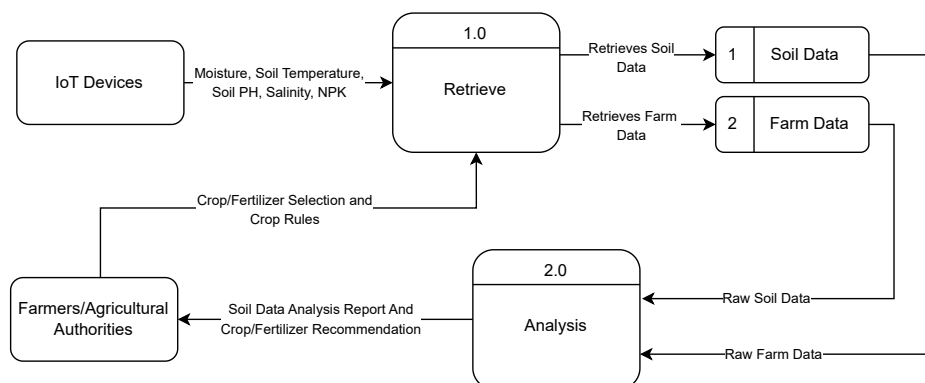


Figure 3.9 shows the Data Flow Diagram (DFD) of the TANIM system. The data flow between the user and the system is divided into two main processes which is the retrieval and analysis process. The “retrieve” process is responsible for collecting soil and farm data from IoT devices,. This includes specific readings of NpH, moisture, salinity, and temperature, as well as crop and fertilizer inputs provided by the users. These data are validated, preprocessed, and stored in the system’s database to ensure consistency, accuracy, and ease of access for subsequent processing. For moisture, temperature, soil pH, the system applies machine learning algorithms. In the Cache Storage, the data flow branches into two distinct paths to evaluate the stored data, identifying patterns and relationships among soil properties, environmental factors to serve the

end users. First, it directly supplies a Soil Data Analysis Report to the Farmers, and crop performance. The outputs of this analysis include predictions on soil health to Agricultural Authorities. Second, it feeds “Real-time Metrics” into the system to monitor crop suitability, and potential yield. Based on these predictions, the system generates tailored crop and fertilizer recommendations that consider both environmental conditions and expected productivity, enabling farmers to make informed decisions using these real-time metrics to generate actionable agricultural insights. The direct output of this process is a tailored decision that optimizes crop and fertilizer resource utilization.

Finally, the processed results are delivered to farmers and agricultural authorities through a user-friendly interface in the form of soil data analysis reports, visual dashboards, and actionable recommendations. These outputs facilitate practical decision-making, support strategic crop planning, and contribute to more efficient and sustainable agricultural practices across Region 10.

Use Case Diagram The Use Case Diagram illustrates how different actors interact with the TANIM system. It presents the system’s primary functionalities and shows how users access and utilize agricultural data for farm management and monitoring. The diagram emphasizes the roles of both Farmers and the Department of Agriculture in supporting efficient and data-driven agricultural practices.

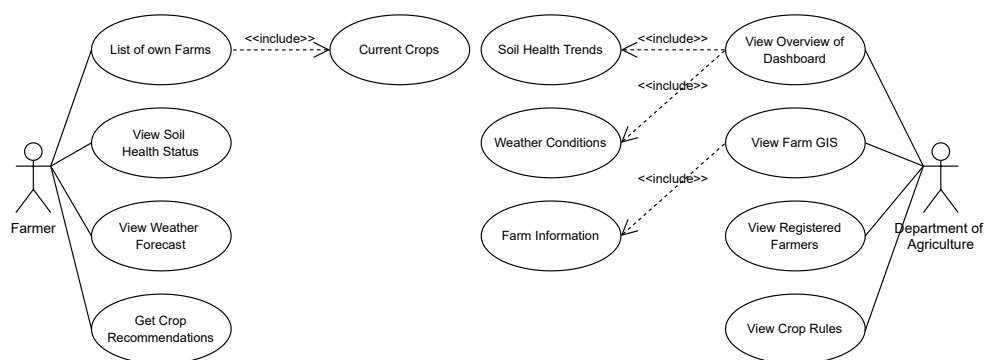
Figure 3.16*Use Case Diagram*

Figure 3.16 shows the interactions of the Farmer with the system. Farmers can view a list of their own farms, which includes information about the current crops being cultivated. This feature helps farmers organize and monitor their farming activities. Farmers are also able to view the soil health status of their farms, allowing them to assess land condition and make informed decisions regarding crop management. In addition, the system provides access to weather forecasts, enabling farmers to plan daily operations and prepare for potential weather-related risks. The system also offers crop recommendations tailored to soil health data, weather conditions, and existing farm information to support improved agricultural productivity.

The figure also illustrates the interactions of the Department of Agriculture with the system. The Department can access an overview dashboard

that presents soil health trends and weather conditions, providing insights into agricultural conditions across different farms. Farm information is integrated into the dashboard to support monitoring and analysis. The Department can also view farm locations through the Farm GIS feature, enabling spatial visualization of agricultural areas. Furthermore, the system allows the Department to view registered farmers and oversee crop rules and regulations, ensuring compliance with established agricultural policies and standards.

The Use Case Diagram demonstrates how TANIM supports both farm-level management and institutional oversight. By providing relevant tools and information to different users, the system enhances coordination, monitoring, and decision-making in agricultural management.

3.5.5 Use Case Table

The Use Case Table provides a structured and detailed representation of the functionalities presented in the Use Case Diagram. It enumerates each system function, identifies the primary actor involved, and describes the purpose of each interaction. In addition, the table highlights relationships among use cases through included or related functionalities.

The use cases associated with the *Farmer* focus on farm-level management and decision support. These include viewing a list of registered farms

and current crops, monitoring soil health status, accessing weather forecasts, and receiving crop recommendations. These functionalities enable farmers to effectively manage their farming activities, assess environmental conditions, and make informed decisions to improve agricultural productivity. Certain use cases depend on related system features, such as soil health and weather data, which support the generation of crop recommendations.

The use cases associated with the *Department of Agriculture* emphasize monitoring, analysis, and regulatory oversight. These include viewing an overview dashboard that integrates soil health trends, weather conditions, and farm information, accessing farm locations through the Farm GIS feature, and viewing registered farmers. The Department may also view crop rules to ensure compliance with agricultural standards and policies.

Table 5: Primary Use Cases of the TANIM System

Use Case Name	Description	Primary Actor	
		Farmer	Department of Agriculture
List Own Farms	Enables farmers to view their registered farms.	✓	
View Soil Health Status	Allows users to view soil condition data such as nutrients, fertility, and health indicators.	✓	
View Weather Forecast	Provides weather forecasts relevant to farm locations.	✓	
Get Crop Recommendations	Provides crop recommendations based on soil health, weather, and farm data.	✓	
View Dashboard Overview	Allows users to view a summarized overview of agricultural data for monitoring and analysis.		✓
View Farm GIS	Displays geographic locations of farms through a GIS interface.		✓
View Registered Farmers	Allows the Department of Agriculture to view all registered farmers in the system.		✓
View Crop Rules	Allows the Department to view and manage crop rules and regulations.		✓

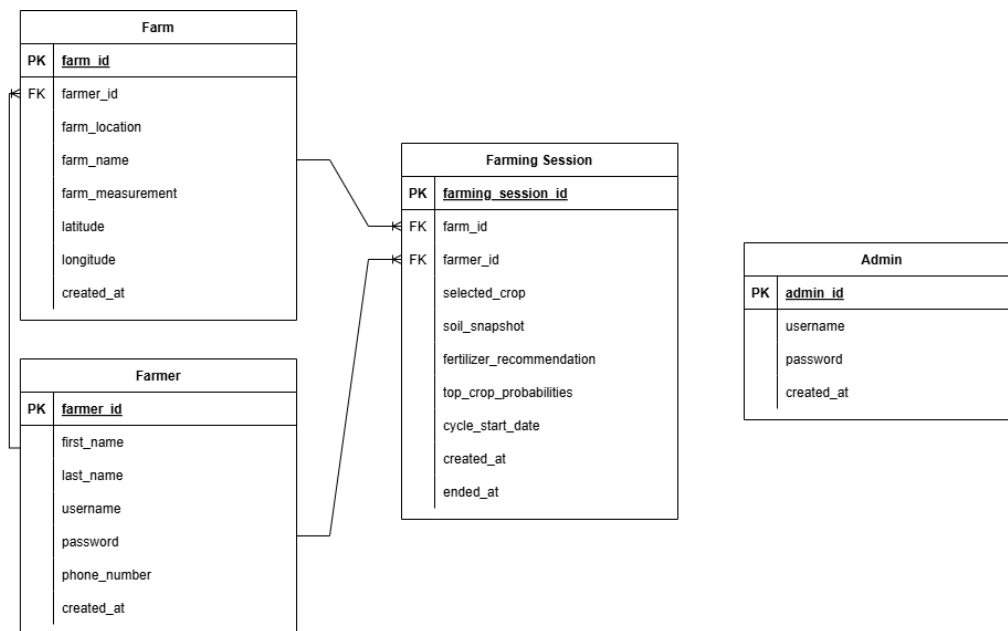
Table 6: Included Use Cases of the TANIM System

Use Case Name	Description	Primary Actor	
		Farmer	Department of Agriculture
View Current Crops	Allows farmers to view crops currently cultivated on their farms.	✓	
View Soil Health Trends	Enables monitoring of soil health changes over time across regions.		✓
View Weather Conditions	Allows the Department to view current and historical weather data.		✓
View Farm Information	Enables the Department to access detailed information about farms.		✓

3.5.6 Entity Relationship Diagram

The relational architecture of the database is designed to provide a structured and scalable environment for managing agricultural data through four primary entities: **Admin**, **Farmer**, **Farm**, and **Farming Session**. As

illustrated in **Figure 3.17**, the **Admin** entity serves as the administrative oversight layer, storing unique identification and authentication credentials for system-wide management. The **Farmer** entity represents the primary end-users of the platform, containing detailed profile information including full names, contact numbers, and secure login credentials. This entity is linked to the **Farm** table through a foreign key (*farmer_id*), establishing a one-to-many relationship that allows a single farmer to register and monitor multiple distinct agricultural plots. Each farm record preserves critical metadata such as the plot name, physical area measurement, and precise geographic coordinates (latitude and longitude) to support spatial mapping.

Figure 3.17*Entity Relationship Diagram*

Operational data and model-driven insights are managed within the **Farming Session** entity, which functions as a temporal log of field activity. This table acts as the nexus of the system, linking a specific farmer to a specific farm for the duration of a planting cycle. Rather than storing soil data as static attributes of a land plot, the system archives discrete “soil snapshots,” fertilizer recommendations, and model-generated top crop probabilities within the session record. This design allows the system to maintain a historical archive of soil health and decision-making for every visit or cycle. The session record also captures the user’s final decision in the *selected_crop* field and

tracks the timing of the cycle through start, end, and creation timestamps. This normalized structure ensures data integrity while supporting complex longitudinal queries regarding farm performance and historical soil trends.

3.5.7 Hardware Design Flowchart

Figure 3.18

Hardware Design Flowchart

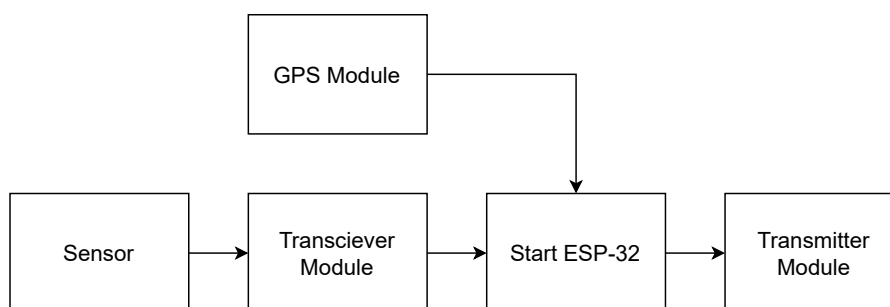


Figure \ref{fig:HardwareFlowchart} illustrates the workflow of the hardware components. The process begins with the initialization of the ESP32 microcontroller, which serves as the central control unit of the system. The GPS module provides location data to the ESP32, while the sensor module collects environmental data required for monitoring.

Once the sensor data is acquired, it is passed to the transceiver module, which facilitates communication between the sensing unit and the ESP32. The

ESP32 processes the received data and forwards it to the transmitter module. Finally, the transmitter module sends the processed data to the designated external system, completing the hardware operation cycle.

The flowchart represents a sequential hardware process in which each stage depends on the successful execution of the previous step. The ESP32 coordinates sensor data acquisition, GPS integration, and communication control, while the transceiver and transmitter modules ensure reliable data transmission.

3.5.8 Hardware Design Layout

3.5.9 3D Design

Figure 3.19

3D Container Design

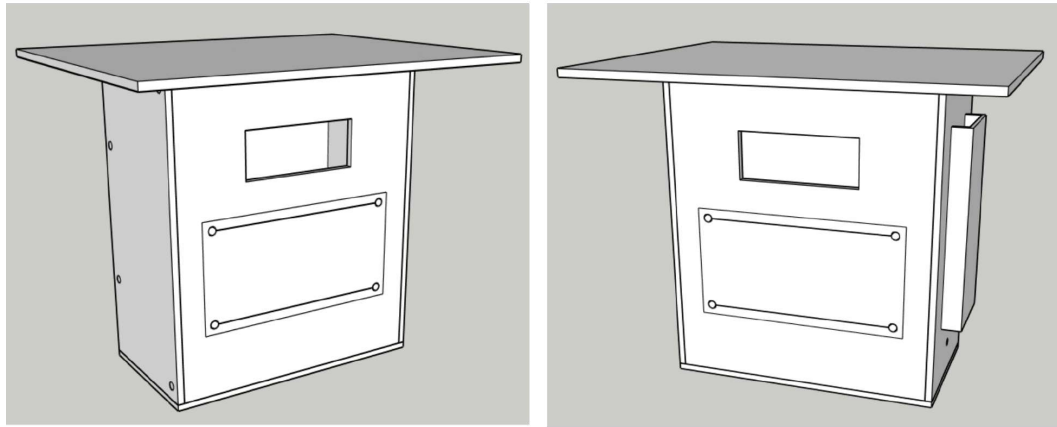


Figure 3.19 displays the 3D container design that will be 3D printed for the container of the hardware. This will contain the hardware components which are etched in a PCB board that will be used in this system. The container should be durable to ensure that it can secure the hardware components inside of it.

3.5.10 Materials and Cost

Table 7: Hardware Components and Cost

Component	Price
ESP32 Microprocessor	P150
Soil Sensor	P1,654.90
Solar Power Manager	P860
Solar Panel	P320
Lithium Ion Battery	P120
GPS Module	P255
Transceiver Module	P163
Transmitter Module	P180
Miscellaneous (Cable, Mounts)	P1,500
Total	P5,202.90

3.6 Implementation

The Implementation phase involves developing the software and hardware components. Each function was ensured to function according to the design specifications. This phase intends to transform the conceptual design into a working prototype that can perform data acquisition, processing, and output for TANIM.

3.6.1 Software Implementation

The software implementation for TANIM is designed to provide an efficient, reliable, and user-friendly platform that caters to the needs of both the Farmer and the Department of Agriculture. To ensure optimal accessibility and performance, the frontend for farmers is developed using React Native, a

framework well-suited for creating responsive and cross-platform mobile applications. The mobile application also features offline data handling, allowing farmers to access previously synced soil and crop data even without an internet connection. Any new inputs or updates made while offline are temporarily stored locally on the device and automatically synchronized with the cloud once connectivity is restored. This ensures uninterrupted functionality in rural areas with limited network coverage.

Meanwhile, the frontend for the Department of Agriculture is implemented using React.js, providing a seamless and interactive web application experience. The web application integrates advanced analytics and visualization tools, enabling agricultural authorities to interpret soil data, monitor farm conditions, and analyze regional agricultural trends through interactive charts, dashboards, and geospatial maps. These insights support data-driven planning, resource allocation, and policy formulation.

Both user interfaces interact with a shared backend developed in Flask, which centralizes data management and ensures consistent functionality across platforms. This architecture enables efficient communication between the mobile and web applications while supporting scalability, maintainability, and ease of integration with additional features or services in the future.

3.6.2 Hardware Implementation

The hardware implementation of the TANIM system is centered on an IoT-based device designed to collect soil data and send it to the cloud for processing and analysis. The core of the device is the ESP32 microcontroller, selected for its high processing capability, built-in Wi-Fi and Bluetooth connectivity, low power consumption, and compatibility with multiple sensor modules. These features make it ideal for continuous field monitoring and data transmission in remote agricultural areas.

A soil sensor is used to measure key soil parameters such as pH, NPK, moisture, temperature, and salinity. This sensor was chosen because it provides accurate and real-time soil data essential for evaluating soil health and guiding fertilizer and crop recommendations.

3.6.3 Machine Learning Model

The TANIM system will use a boosting algorithm as the core of its predictive model, chosen for its efficiency, scalability, and high accuracy in handling large and complex datasets. This machine learning approach analyzes multiple factors, including soil type, environmental conditions, and historical yield data, to generate accurate predictions of soil health, crop suitability, and potential yield. By leveraging the predictive power of boosting

algorithms, the system is able to deliver personalized and data-driven crop and fertilizer capabilities of LightGBM with advanced synthetic data generation, the system outputs precise, site-specific agronomic recommendations that are designed to optimize productivity and promote efficient resource utilization to yield and maximize resource use efficiency. This will be strictly enforced at the level of a Farmer cannot accidentally or maliciously access DA-level administrative rules. Two-Factor Authentication (2FA) will also be implemented during the login process for both Farmers and Agriculturalists in the mobile or web application, users will be required to provide a secondary verification code sent to their registered email or mobile number. This adds a critical layer of protection against unauthorized access to sensitive farm data and regional soil health insights.

The TANIM system secures the exchange of information across multiple layers of the architecture. For the long-range link between the field sensors and the gateway, AES-128 encryption is a standard feature of LoRa technology that is utilized to ensure that soil parameters remain confidential during wireless transmission. Furthermore, communication between the Flask backend and the user applications is conducted over HTTPS (TLS/SSL) to prevent data interception or man-in-the-middle attacks. This ensures that real-time insights, such as fertilizer recommendations and crop suitability scores, are

delivered securely to the end-user.

Data integrity begins at the collection point, where the ESP32 micro-controller handles localized processing and filtering to prevent signal spikes from being recorded as valid data. The hardware is protected by a durable 3D-printed container, designed to secure the PCB and sensors from physical tampering in remote agricultural settings. On the cloud side, Firebase Security Rules are configured to ensure data isolation; Farmers are only permitted to write data associated with their specific accounts, while the Department of Agriculture is granted read-only access to regional soil health trends for monitoring purposes. Regular database backups and encryption-at-rest further safeguard the repository of environmental and farmer-defined parameters.

For wireless communication, the system uses a LoRa (Long Range) module, selected for its ability to transmit data over several kilometers with very low power consumption. This makes it suitable for rural agricultural zones with limited internet access or unstable connectivity.

The entire system is powered by a solar panel managed through a solar panel controller, ensuring a sustainable and uninterrupted power source even in off-grid farming locations. The modular design of the hardware also makes the system scalable and adaptable, allowing additional sensors or communication modules to be integrated easily as agricultural requirements evolve.

3.7 Test and Evaluation

Testing and evaluation will be conducted to verify the system's effectiveness, accuracy, and usability for crop recommendation and soil health monitoring. This process will be designed to ensure that the hardware components, software components, and machine learning models will function according to what we aim for this study, which involves hardware testing and calibration, functional testing, system integration testing, performance testing, and usability testing. Hardware testing and calibration will focus on verifying the accuracy of the sensors to ensure reliable soil data. Functional testing will examine whether the mobile and web applications operate correctly and deliver the expected outputs of each feature. To ensure that the hardware, cloud database, machine learning, and applications works together with proper data flow and error handling, system integration will be implemented. For measuring the responsiveness of the system and to evaluate the system's predictive accuracy of the machine learning models using various performance metrics we will be conducting performance testing. Moreover, usability testing will assess the practicality and user-friendliness of the system by involving farmers and agricultural experts in field trial and feedback collection.

3.7.1 Hardware Testing and Calibration

Hardware testing and calibration will ensure that the IoT sensors will provide accurate measurements of soil health parameters such as nitrogen, phosphorus, potassium, pH, moisture, salinity, and temperature. To conduct this testing, the sensors will be deployed in field testing on the actual farms in Region 10, also to ensure consistency under varying field conditions the readings will be recorded at different time and day. Cross checking will be applied by using the Department of Agriculture's soil testing laboratory and soil test kits of the same soil samples from the same farms to ensure reliability and accuracy.

Through statistical measures such as Mean Absolute Error (MAE) the sensor readings will be compared against laboratory results. a maximum allowable error threshold of ± 5 will be set, if the difference between the sensor and laboratory values exceeds this threshold error, calibration adjustments will be applied to the sensors until acceptable accuracy will be achieved. It will be calculated by:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

This process ensures that the data generated by the hardware is reliable enough to be used in machine learning models and the recommendation

system.

3.7.2 Functional Testing

Functional testing involves verifying whether each feature of the mobile and web applications operates correctly according to the system design. For the mobile application, the following features will be tested: secure login and authentication, retrieval of soil health readings, display of crop recommendations for multi-cropping, fertilizer optimization, and offline data synchronization. For the web application, features such as farm visualization, farmer profile management, and crop rule configuration will be tested.

In addition, synchronization between the database, mobile, and web applications will be tested to ensure that all data are accurately updated and consistent across platforms. This includes verifying that soil readings, user information, and crop recommendations are properly synchronized when data are sent or retrieved from the cloud.

Both black-box testing and white-box testing approaches will be applied to evaluate the system's functionality. Black-box testing will focus on verifying the system's external behavior without examining its internal code. Testers will input various data to check if the outputs match the expected results, ensuring that all functions operate as intended from a user's perspec-

tive. Meanwhile, white-box testing will be conducted by examining the internal logic, data flow, and code structure of the applications. This will help identify hidden logical errors, broken functions, or inefficient code implementations.

Stress testing will also be performed by simulating multiple user logins and concurrent farm entries to evaluate system performance under heavy load. Through these testing activities, the TANIM system aims to ensure accurate functionality, reliable synchronization between components, and stable performance in real-world agricultural applications.

3.7.3 System Integration Testing

System integration testing will evaluate whether the hardware, software, cloud database, and machine learning model communicates seamlessly with each other as a single system. The system integration testing will be tested by following the full flow of the system: soil health readings captured by the sensor and processed by the ESP32 microcontroller, then transmitted by via the LoRa module to the gateway, forwarded to the cloud database, then processed by the machine learning models, and finally displayed on the mobile and web applications.

This process confirms that the system works together as one reliable system. Furthermore, tests will be conducted to observe how the system han-

dles incomplete or corrupted data to ensure that it can manage error to prevent misleading recommendations to farmers.

3.7.4 Performance Testing

For the system level performance, it is to determine if the system could handle multiple data acquisition and provide timely feedback to the users. The data transmission success rate will be monitored by comparing the successful sensor to cloud uploads against the attempted uploads over multiple trials. Latency tests measure the time from the sensor data capture to its display in the mobile and web application, with an acceptable range set to 30 seconds. Scalability will be evaluated by simulating multiple farms connected simultaneously to ensure that the cloud database and applications could handle increased workloads without delays and crashes. These will ensure that the system is reliable and efficient under real-world agricultural conditions where multiple users operate.

For the machine learning evaluation, it verifies the predictive capability of the machine learning models that will be used on crop recommendation for multi-cropping and fertilizer optimization. The dataset will be split into 70-15-15 for training, evaluation, and testing for maximizing the predictive accuracy and avoiding overfitting.

For the crop recommendation, the machine learning models will be evaluated using accuracy, precision, recall, F1-score, and confusion matrices to determine the effectiveness of the system. The following formulas are going to be use:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

$$\text{Confusion Matrix} = \begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix} \quad (6)$$

These evaluation steps confirm that the machine learning models provide scientifically sound predictions and deliver crop and fertilizer recommendations that are consistent, reliable, and effective under different scenarios.

3.7.5 Usability Testing

The usability testing will assess the practicality, accessibility, and user satisfaction of the system. For this, farmers will be asked to log into the mobile application, review soil health data, view crop and fertilizer recommendations, and simulate updating their farm records. Meanwhile, DA officials will use the web dashboard to access farmer profiles, view GIS-based maps, and manage crop rules.

A total of 10 participants will be involved in the usability testing, consisting of 7 farmers from Region 10 and 3 agricultural technicians or DA employees. Farmers were selected based on their familiarity with basic smartphone usage and their active involvement in small- to medium-scale farming. The DA personnel were chosen because they represent the system's administrative users responsible for monitoring and validating agricultural data. This mix of participants ensures that both end-user and administrative perspectives are considered during evaluation.

After completing the navigation of the system, the System Usability

Scale (SUS) developed by Brooke (1995) will be employed to quantitatively measure user satisfaction. The interactions of the users will be observed, and any difficulties they encounter will be recorded. In addition to the SUS scores, qualitative feedback will be collected through short interviews and observation notes to gain insights into which features are most useful and which aspects require improvement. The evaluation criteria will determine the user satisfaction based on the SUS scores and user feedback. The result of this testing will determine the system's usability rating and this will be a guide for further refinement to ensure that the TANIM system is accessible, user-friendly, and beneficial to the users.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the findings from the research conducted and provides a thorough analysis and interpretation of these results.

4.1 Sensor Hardware Testing and Calibration

This section presents the results of the sensor hardware testing and calibration conducted to evaluate the accuracy, reliability, and consistency of the developed IoT-based soil sensing unit. As described in Chapter 3, the system measures key soil parameters including pH, nitrogen (N), phosphorus (P), potassium (K), soil moisture, temperature, and salinity. The accuracy of these parameters is critical since they directly influence the crop recommendation and fertilizer advisory outputs of the system.

The testing phase aimed to (1) verify the correctness of sensor readings, (2) calibrate the sensors against reference values, and (3) evaluate the performance of the sensing unit under controlled conditions. The results obtained from this phase serve as the foundation for validating the reliability of the entire system.

4.1.1 Testing Setup and Environment

4.1.2 Sensor Calibration Procedure

4.1.3 Accuracy Evaluation Metrics

4.1.4 Results and Analysis

4.1.5 Discussion of Findings

4.2 Data Processing and Preparation Results

This section presents the results of the data processing and preparation procedures applied to the collected datasets. Multiple data sources such as soil sensor readings, environmental data, zoning information, and historical agricultural records were consolidated and transformed into a structured format suitable for analysis.

The primary objective of this phase was to ensure that the dataset is accurate, consistent, and representative of real-world agricultural conditions in Bukidnon. Various techniques such as data integration, cleaning, preprocessing, and feature engineering were applied to improve data quality and usability.

The results discussed in this section demonstrate how raw and heterogeneous data were refined into a reliable dataset that supports effective machine learning modeling and accurate crop and fertilizer recommendations.

4.2.1 Data Integration Results

The data integration phase yielded a structured and machine-readable dataset derived from the Consolidated Soil Info (RSL) 2023–2025 repository. As illustrated in **Figure 4.1**, the finalized dataset consists of validated soil profiles defined by key agricultural parameters: soil pH, organic matter per-

centage, and nutrient concentrations of phosphorus and potassium in parts per million (ppm). Following the integration process, irrelevant metadata and unstructured annotations were removed to produce a clean dataset centered on agronomic variables. The processed data was subsequently mapped to a predefined set of ten target crops—including Corn, Tomato, and Cassava—ensuring categorical consistency across the repository.

Figure 4.1
Sample Integrated Soil Data Records

Sample Integrated Soil Data Records					
Record ID	Soil pH	Organic Matter (%)	Phosphorus (ppm)	Potassium (ppm)	Zoned Target Crop
RSL-001	7.22	2.80	34.60	47.00	Corn
RSL-002	5.09	4.70	1.00	121.00	Camote (Sweet Potato)
RSL-003	5.09	4.70	1.00	121.00	Corn
RSL-004	5.09	4.70	1.00	121.00	Eggplant
RSL-005	5.09	4.70	1.00	121.00	Tomato

The resulting dataset is composed of structured entries where each row represents a complete soil profile linked to a specific crop classification. For instance, entries such as RSL-001 through RSL-005 in **Figure 4.1** demonstrate the alignment of specific chemical properties with zoned target crops like Corn and Eggplant. This structural transformation converted fragmented spreadsheet data into a uniform format suitable for machine learning model

training. Ultimately, the integration phase produced a reliable baseline that bridges historical agricultural records with the data structures required for real-time Internet of Things (IoT) sensor simulation.

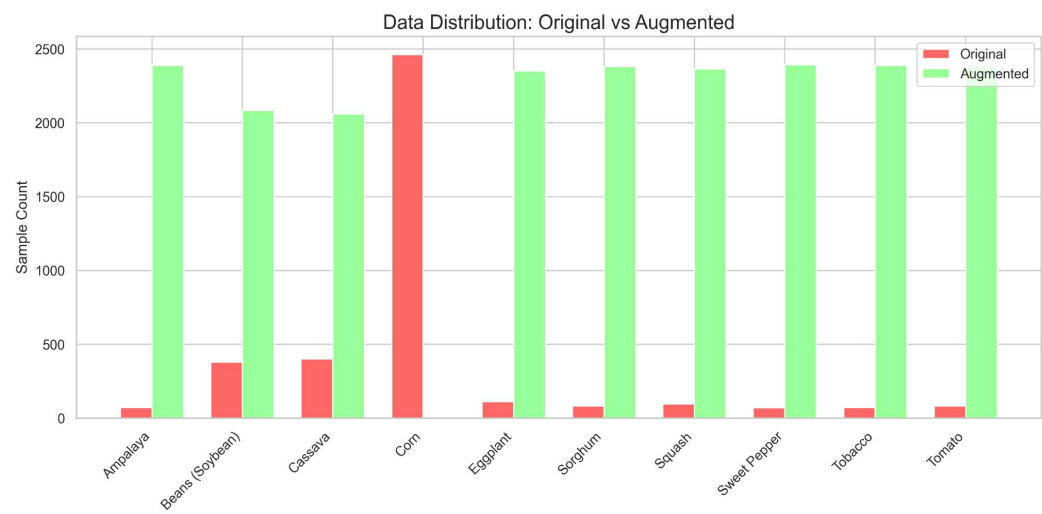
4.2.2 Data Cleaning and Preprocessing Results

The preprocessing stage resulted in a significantly improved dataset in terms of completeness, consistency, and class distribution. All missing and invalid entries, including non-numeric errors and incomplete sensor readings, were successfully eliminated, producing a fully clean dataset with no null or corrupted values. This rigorous cleaning process ensured that the secondary data ingested from historical records remained structurally consistent with the real-time telemetry captured by the sensor node.

Noise and outlier reduction further enhanced the dataset quality. All numerical features were constrained within realistic agricultural limits, ensuring physically valid values across all parameters. For instance, soil pH values were maintained strictly within the acceptable agronomic range of 3 to 9, preventing the presence of unrealistic or extreme measurements that could skew model convergence. This normalization resulted in a more stable and reliable data distribution, effectively preparing the feature space for gradient-based learning.

A major improvement was observed in the class distribution of the target variables. Prior to preprocessing, the dataset exhibited severe class imbalance, as shown in the original (red) bars in **Figure 4.2**, where common crops like Corn significantly outnumbered minority categories such as Ampalaya and Sweet Pepper. This disparity posed a high risk of producing a biased model that favored majority class predictions. After the application of synthetic data balancing, the dataset achieved a uniform class distribution, as illustrated by the augmented (green) bars in **Figure 4.2**.

Figure 4.2
Data Distribution Comparison Between Original and Augmented Datasets



Note. The figure demonstrates the resolution of class imbalances through synthetic augmentation, resulting in a uniform sample count of approximately 2,400 across all target crops.

The resulting dataset reached a balanced state with approximately

2,400 samples per crop category, providing a statistically stable foundation that improves the reliability and generalization of the subsequent machine learning training. Overall, the preprocessing phase successfully transformed the raw inputs into a clean, balanced, and statistically stable dataset, effectively eliminating sources of bias and ensuring the integrity of the predictive engine.

4.2.3 Feature Engineering Outcomes

The feature engineering phase resulted in the expansion of the dataset from four original soil properties to eight total predictive variables through the derivation of four new features grounded in established agronomic relationships. These algorithmically derived features, summarized in **Table 8**, mirror physiological nutrient constraints and bio-availability dynamics that are critical for accurate soil-health assessment.

Table 8: Feature Engineering Outcomes and Agronomic Justifications

Engineered Feature Name	Mathematical Formulation	Agronomic Justification
Phosphorus-Potassium Ratio	$\frac{\text{Phosphorus (ppm)}}{\text{Potassium (ppm)} + 1}$	Evaluates the internal root development-to-fruiting energy balance. A severe imbalance indicates a localized deficiency incapable of supporting advanced botanical maturity.
Phosphorus-Organic Load	$\frac{\text{Phosphorus (ppm)}}{\text{Organic Matter (\%)} + 1}$	Quantifies the availability of Phosphorus relative to the baseline organic mass, identifying environments where high raw nutrients are inaccessible due to poor soil degradation.
pH-Gated Nitrogen Bio-availability	$\text{Organic Matter (\%)} \times pH$	Synthesizes an intuition for real-world Nitrogen uptake. Because organic nitrogen conversion is strictly mediated by microbiological activity, this vector encodes active bio-availability.
pH-Gated Potassium Bio-availability	$\text{Potassium (ppm)} \times pH$	Simulates real-world Potassium fixing limitations. In highly acidic environments, absolute Potassium lock-up occurs; this parameter forces the model to acknowledge acidity constraints.

Two ratio-based features were successfully computed: the Phosphorus-to-Potassium ratio (P_K_ratio) and the Phosphorus-to-Organic Load ratio (P_OM_ratio). As detailed in **Table 8**, the P_K_ratio evaluates the internal balance between root development and fruiting energy, while the P_OM_ratio identifies environments where raw nutrients are inaccessible due to poor soil degradation. These features enable the model to evaluate relative nutrient proportions, which often carry more discriminative power in tropical soil profiles than absolute concentrations alone.

Furthermore, two interaction features were generated: the pH-Gated Nitrogen Bio-availability ($OM_pH_interaction$) and the pH-Gated Potassium Bio-availability ($K_pH_interaction$). These variables mathematically encode the dependency of nutrient uptake on soil acidity. Specifically, these terms simulate real-world Potassium fixing limitations and Nitrogen conversion mediated by microbiological activity.

The final set of eight features served as the basis for model training. Feature importance analysis via LightGBM split gain confirmed that the engineered features contributed meaningfully to the system’s predictive capability, collectively accounting for approximately 42.4% of the total feature importance. Notably, the $K_pH_interaction$ feature ranked as the third most important variable overall at 15.1%, surpassing two raw soil measurements. The

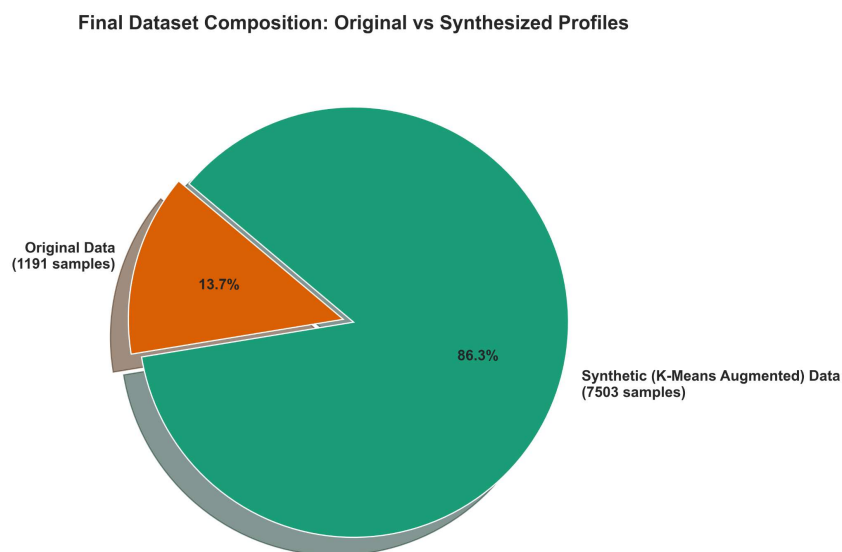
P_{OM_ratio} (10.0%) and P_{K_ratio} (9.3%) further demonstrated that relative nutrient proportions carry substantial power for crop classification. Overall, the feature engineering process produced a richer and more informative dataset that enabled the classifier to capture complex agronomic relationships.

4.2.4 Final Datasets Summary

The final dataset composition represents the culmination of rigorous cleaning, synthetic augmentation, and feature engineering, resulting in a repository optimized for high-fidelity predictive modeling. As illustrated in **Figure 4.3**, the K-Means augmentation protocol substantially expanded the data volume from an initial 1,191 original samples (13.7%) to include 7,503 synthetic profiles (86.3%). This expansion ensures that the crop classes are represented with statistical parity, effectively resolving the severe class imbalances present in the raw historical records.

Figure 4.3

Final Dataset Composition: Original vs. Synthesized Profiles



Each of the 8,694 total records is defined by eight predictive features: the four primary parameters targeted for real-time sensor acquisition—Organic Matter, Phosphorus, Potassium, and pH—and the four mathematically derived agronomic interaction variables. Before being transitioned to the machine learning workflow, the dataset underwent a final standardization to neutralize scalar differences between chemical measurements and was encoded to transform categorical crop labels into discrete integers. This finalized, mathematically uniform environment is structurally prepared to train a robust classification model capable of interpreting active field telemetry within the TANIM framework.

4.3 Machine Learning Model Performance

This section presents the performance of the machine learning model used for crop recommendation based on soil health and environmental parameters. The evaluation focuses on how well the model predicts suitable crops using the processed dataset.

The model was assessed using standard classification metrics to determine its accuracy, consistency, and reliability. The results provide insight into the effectiveness of the model in identifying patterns within the dataset and generating accurate predictions.

4.3.1 Model Configuration

The TANIM crop recommendation system utilizes a Light Gradient Boosting Machine (LightGBM) classifier as its predictive engine. The selection of this algorithm was based on its computational efficiency, rapid execution speed, and robust capability to model nonlinear interactions inherent in complex agricultural datasets. By employing a gradient boosting framework that constructs decision trees leaf-wise rather than depth-wise, the algorithm achieves maximized classification accuracy while simultaneously minimizing memory overhead. This represents a critical architectural advantage for subsequent integration into resource-constrained Internet of Things (IoT) deploy-

ments. Functioning as a robust multiclass classifier, the system is designed to map environmental parameters collected in real time to the most agronomically suitable crop from a predefined selection.

The baseline input features ingested by the model are derived from critical soil physicochemical properties continuously collected via the sensor network. These include Organic Matter, expressed as a percentage and functioning as a proxy for Nitrogen; Phosphorus, measured in parts per million (ppm); Potassium, also in ppm; and soil pH levels. To augment the predictive depth of the system, a dedicated feature engineering pipeline introduces four derived agronomic features explicitly formulated to capture vital soil and crop interaction effects. These engineered variables encompass the Phosphorus-to-Potassium ratio (P/K), the Phosphorus-to-Organic Matter ratio (P/OM), and mathematical interaction terms between pH and nutrient availability, specifically $Organic\ Matter \times pH$ and $Potassium \times pH$. By synthesizing this set of eight total features, the model is equipped to discern profound agronomic phenomena, such as the manner in which pH variations modulate the physiological uptake of localized nutrients.

To ensure representational fairness and mitigate predictive bias, a rigorous data preprocessing pipeline has been implemented. Empirical agricultural datasets frequently exhibit class imbalances, where the profiles of certain

prevalent crop varieties substantially outnumber those of minority varieties. To address this issue, a synthetic data augmentation technique utilizing K-Means Clustering is applied. This methodology identifies dimensional clusters within the minority classes and generates targeted localized synthetic samples infused with Gaussian noise, thereby achieving a smooth balancing of the dataset. Subsequent to augmentation, features undergo standardization, and the categorical crop targets are mathematically transformed utilizing label encoding.

Rather than relying on static heuristic parameters, the architecture of the classifier was systematically optimized using Bayesian hyperparameter optimization facilitated by the Optuna framework. The optimization process involved the execution of 50 independent trials evaluated under a rigorous five-fold stratified cross-validation protocol. This protocol ensured that the proportional representation of each crop class remained perfectly balanced across all training and holdout evaluation folds. The Optuna study was configured to minimize multiclass logarithmic loss by dynamically exploring an expansive parameter space. This optimization included parameters governing tree structural depth, specifically *max_depth* and *num_leaves*; convergence pacing through *learning_rate*; stochastic ensemble sampling techniques such as *feature_fraction* and *bagging_fraction* to counteract overfitting; and the

fine-tuning of L1 and L2 regularization penalties, denoted as λ_{l1} and λ_{l2} . The final model weights are a direct product of the parameters yielded by the best-performing trial.

Upon deployment, the model generates a probability distribution across all candidate crop categories, algorithmically isolating the highest-scoring outcome as the primary crop recommendation alongside a calculated prediction confidence percentage. This primary inference seamlessly informs a secondary subsystem responsible for generating actionable crop-specific fertilizer recommendations. These recommendations are strictly based on the discrepancies observed between the current soil state and the ideal agronomic conditions for the chosen crop. Notwithstanding the sophistication of this pipeline, the system operates under specific technical assumptions necessitated by hardware constraints. Principally, standard commercial soil sensors often estimate explicit Nitrogen, Phosphorus, and Potassium values indirectly through broad surrogate measurements, such as electrical conductivity. Consequently, these inputs function as highly correlated proxies rather than precise laboratory-grade chemical extractions. The model natively accommodates this hardware limitation by assuming that the measured Organic Matter and resultant electrical conductivity metrics constitute sufficient and consistent correlative indicators for localized soil fertility.

4.3.2 Training and Validation Setup

The foundation of the TANIM crop recommendation framework is based on an empirical dataset obtained from the Consolidated Soil Information (RSL) repository covering the years 2023 to 2025. The dataset includes key soil physicochemical properties such as nitrogen derived from organic matter, phosphorus, potassium, and soil pH, which are associated with historical crop suitability. These parameters serve as the ground truth for training the machine learning model and for validating the data collected from the Internet of Things sensing unit.

To systematically evaluate the predictive performance of the model, the dataset was divided into training, validation, and testing subsets using a 70 percent, 15 percent, and 15 percent split, respectively. A stratified sampling approach was applied to preserve the proportional distribution of crop classes across all subsets. This ensured that each crop category was properly represented during training, tuning, and final evaluation, thereby reducing sampling bias.

A common challenge in agricultural datasets is class imbalance, where widely cultivated crops are overrepresented compared to less common varieties. To address this, a localized oversampling technique based on K-Means clustering was applied to the training dataset. This method groups minor-

ity class samples and generates synthetic data points near cluster centroids using controlled perturbation. This improves class balance while preserving the structure of the original data, allowing the model to learn more effectively across all crop categories.

The model training process utilized a Light Gradient Boosting Machine algorithm, which was optimized using a Bayesian hyperparameter tuning framework implemented through Optuna. The optimization process consisted of 50 trials evaluated using five-fold stratified cross-validation within the training dataset. The validation subset was used to monitor model performance during tuning and to guide the selection of optimal hyperparameters. This approach ensured that the model achieved strong generalization while minimizing overfitting.

During optimization, key parameters such as learning rate, number of leaves, and tree depth were adjusted. Regularization techniques, including L1 and L2 penalties, were applied to control model complexity. In addition, feature sampling and data bagging were incorporated to improve robustness and prevent over-reliance on specific data patterns.

After determining the optimal hyperparameters, the final model was trained using a multiclass logarithmic loss objective function for up to 500 boosting iterations. An early stopping mechanism was implemented based on

validation performance, where training was halted if no improvement was observed over 20 consecutive iterations. This ensured that the model parameters were captured at their optimal state without overfitting.

The final model was evaluated using the unseen 15 percent testing dataset. Performance was measured using multiple evaluation metrics, including accuracy, precision, recall, and F1-score. These metrics provided a comprehensive assessment of the model's predictive capability across all crop classes. The results indicate that the model is capable of generating accurate and balanced crop recommendations, even in the presence of class imbalance within the dataset.

4.3.3 Performance Metrics

To rigorously quantify the operational efficacy of the crop recommendation engine, the Light Gradient Boosting Machine (LightGBM) classifier was evaluated against the unseen holdout subset utilizing a multidimensional array of statistical performance metrics. As illustrated in **Figure 4.4**, the model achieved a global accuracy of 83.2%, providing a robust baseline for its predictive capability. While standard global accuracy offers a general understanding of the overall prediction rate, it can occasionally misrepresent true performance in agricultural datasets. Consequently, balanced accuracy

was prioritized to establish an equitable metric, functionally calculating the macro-average recall across each crop category. This approach ensures that synthetically augmented minority classes are predicted with the same fidelity as natively dominant ones, maintaining the integrity of the recommendation engine across all ten targeted crops.

Figure 4.4
Model Classification Report: Precision, Recall, and F1-Score Breakdown

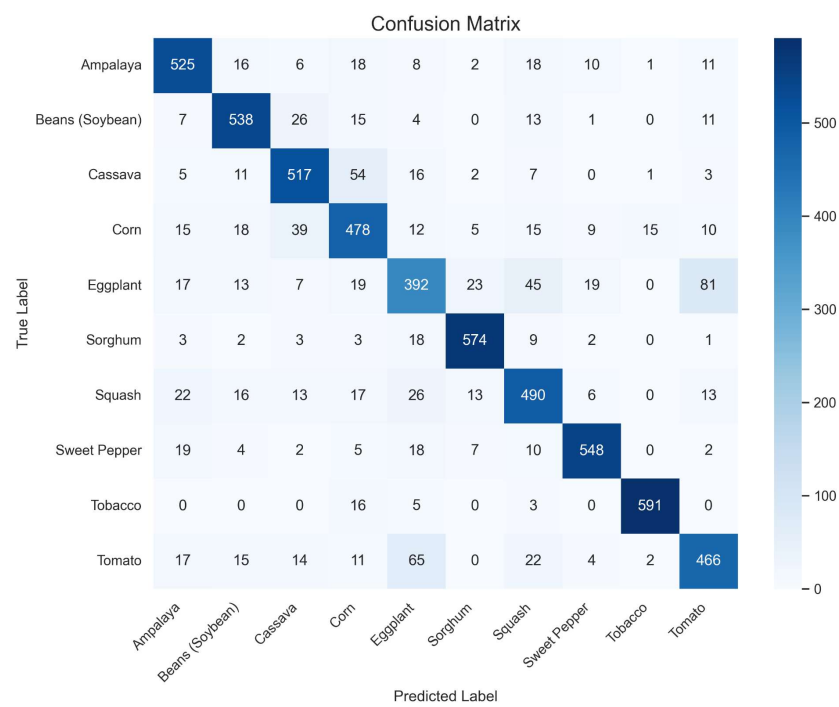
MODEL CLASSIFICATION REPORT			
Precision, Recall, and F1-Score Breakdown			
	precision	recall	f1-score
Ampalaya	0.833	0.854	0.843
Beans (Soybean)	0.850	0.875	0.862
Cassava	0.825	0.839	0.832
Corn	0.752	0.776	0.764
Eggplant	0.695	0.636	0.664
Sorghum	0.917	0.933	0.925
Squash	0.775	0.795	0.785
Sweet Pepper	0.915	0.891	0.903
Tobacco	0.969	0.961	0.965
Tomato	0.779	0.756	0.768
accuracy	0.832	0.832	0.832
macro avg	0.831	0.832	0.831
weighted avg	0.831	0.832	0.831

To analyze prediction granularity, the evaluation extracted precision, recall, and the resulting F1-score for each crop type. Precision quantified the exact proportion of positive crop recommendations that were agriculturally correct, while recall measured the proportion of actual viable instances that the model successfully identified. These characteristics were harmonized into a weighted F1-score of 0.831, which serves as the definitive harmonic mean of the classifier. As shown in **Figure 4.4**, the model demonstrated exceptional performance for Tobacco, yielding a near-perfect F1-score of 0.965, whereas Eggplant proved more challenging with an F1-score of 0.664. Within the specific operating context of an agricultural Internet of Things (IoT) system, optimizing this F1-score is paramount; it guarantees that the operational and financial resources of the end user are not expended on false positive recommendations while ensuring that highly viable, niche crop opportunities are systematically identified. Furthermore, localized multiclass logarithmic loss was utilized as the mathematical penalty benchmark during the Optuna hyperparameter trials, penalizing the algorithm for false classifications at an intensity proportional to unwarranted model confidence.

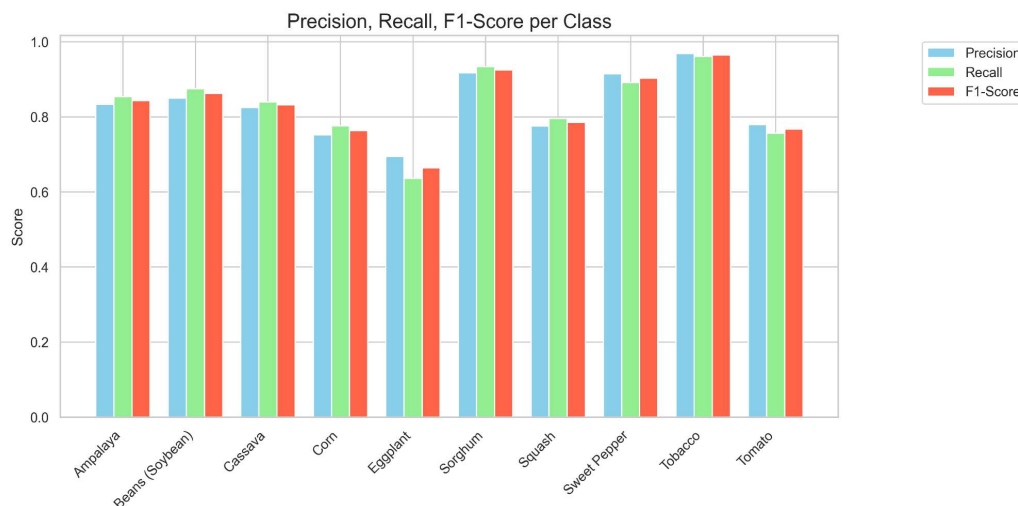
4.3.4 Results Presentation

To provide a clear and diagnostic overview of the classifier, the testing outcomes were synthesized into distinct analytical visual representations. A primary confusion matrix was generated to visualize the model's predictive accuracy against ground truth data across all ten crop categories. As illustrated in **Figure 4.5**, the heatmap reveals high classification fidelity, particularly for Tobacco and Sorghum, where the diagonal values indicate a high volume of correct predictions. Conversely, the matrix highlights specific misclassifications; for instance, Eggplant was occasionally confused with Tomato (81 instances) and Squash (45 instances), likely due to overlapping soil parameter profiles between these varieties.

Figure 4.5
Confusion Matrix of the LightGBM Classifier Across Ten Crop Varieties



To supplement the text-based metrics, granular graphical representations were utilized to evaluate the consistency of the model. **Figure 4.6** presents a per-class breakdown of the precision, recall, and F1-score variables. This bar chart confirms that the predictive success is uniform across most classes, with Tobacco and Sorghum maintaining the highest scores, while identifying the inherent complexity in classifying Eggplant, which exhibits the lowest metrics.

Figure 4.6*Precision, Recall, and F1-Score Breakdown per Crop Category*

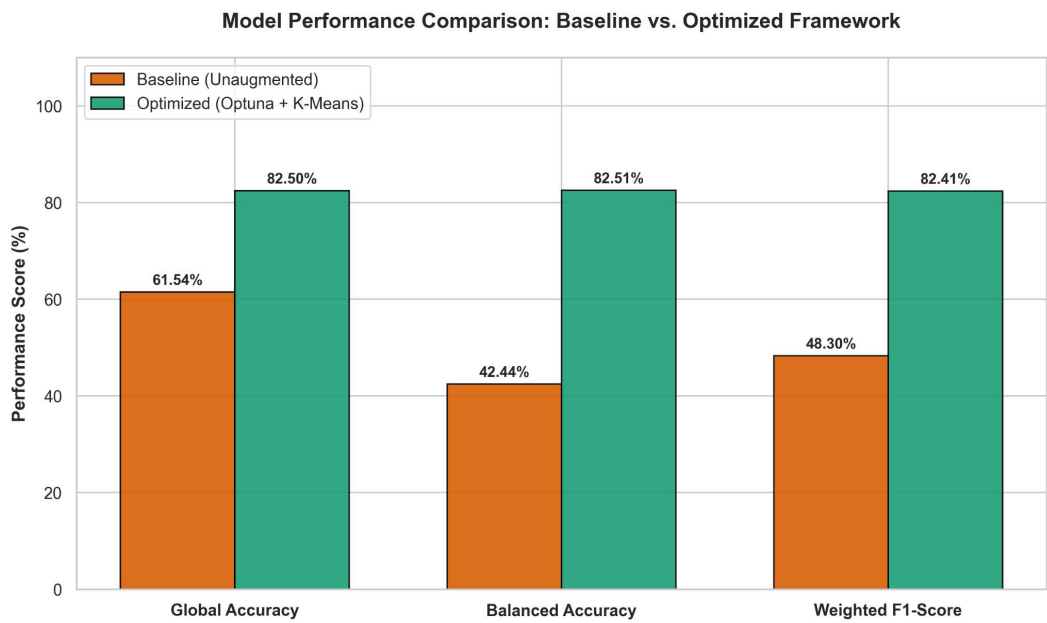
Furthermore, a feature importance analysis was conducted to identify which hardware variables—such as specific soil pH ranges or derived nutrient interaction equations—provided the most significant informational gain for the decision trees. These visual tools, combined with histograms assessing the overall prediction confidence distribution, facilitate a comprehensive understanding of how the system translates raw hardware telemetry into reliable agricultural recommendations.

4.3.5 Discussion of Results

The empirical results derived from executing the Optuna-optimized Light Gradient Boosting Machine (LightGBM) framework alongside K-Means

synthetic augmentation demonstrate a profound operational improvement over standard baseline paradigms. Preliminary computational testing explicitly highlighted that algorithms subjected to raw and unaugmented imbalanced distributions inherently skewed toward majority crops, traditionally yielding unacceptably low and non-viable balanced accuracies. As illustrated in **Figure 4.7**, the integration of aggressive hyperparameter optimization and synthetic Gaussian interpolation of minority classes definitively neutralized this systemic bias, elevating balanced accuracy from 42.44% in the baseline model to 82.51% in the optimized framework.

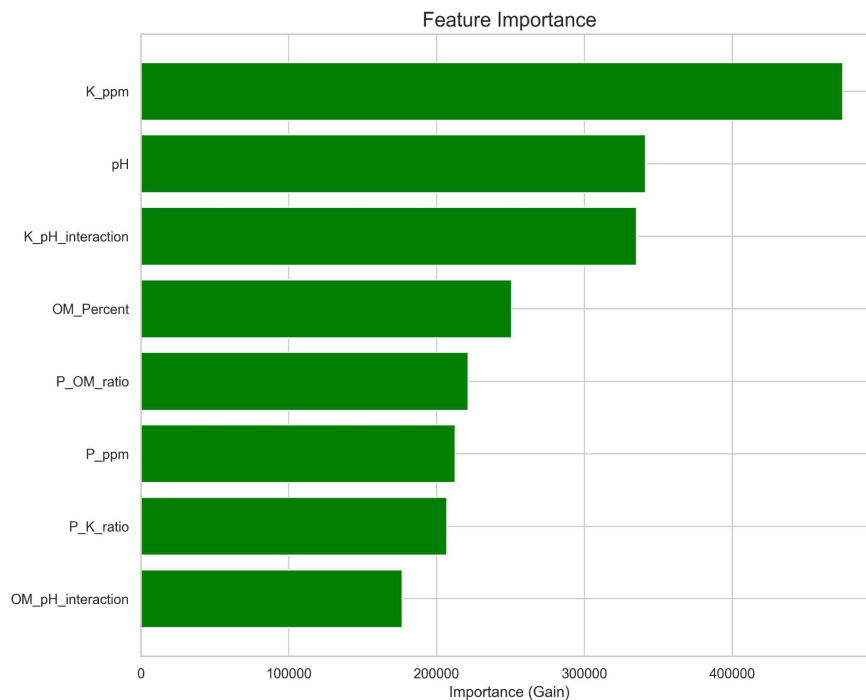
Figure 4.7
Model Performance Comparison: Baseline vs. Optimized Framework



Furthermore, the explicit analysis of the feature importance extraction serves to validate the foundational hardware architecture characterizing the underlying Internet of Things (IoT) system. The evaluation presented in **Figure 4.8** confirms that core attributes, specifically Potassium (K_ppm), localized soil pH, and the intentionally engineered $K_pH_interaction$ term, act as the most deterministic mathematical indicators for overarching crop viability.

Figure 4.8

Feature Importance Analysis Measured by Information Gain

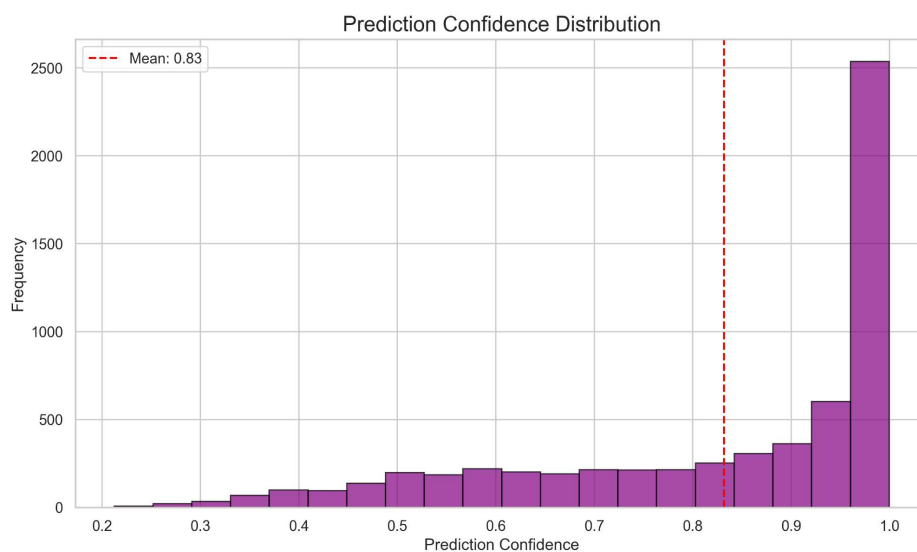


Crucially, the prediction confidence distribution visually verifies that

the vast majority of the inferences made by the classifier are anchored at high certainty probability thresholds rather than marginal estimations. As shown in **Figure 4.9**, the model maintains a mean confidence of 0.83, with a significant concentration of predictions reaching the 0.9 to 1.0 range.

Figure 4.9

Prediction Confidence Distribution of the Optimized Classifier



For the end-user managing an automated agricultural deployment, these results confirm that the integrated software functions natively as a robust decision-support tool. The pipeline is mathematically equipped to reliably identify optimized cultivation pathways and actively reject suboptimal environmental variables with measurable statistical certainty. Ultimately, this structural reliability ensures that any subsequent downstream sequences exe-

cuted by the system, such as computing precise and dynamic fertilizer application schedules, are grounded upon recommendations that are consistently, dependably, and agronomically sound.

4.4 Crop Recommendation Validation

This section presents the validation of the crop recommendation outputs generated by the system. The purpose of this evaluation is to determine whether the recommendations produced by the machine learning model are accurate, reliable, and aligned with expected agricultural practices.

The validation process involved comparing the system's predicted crop recommendations with reference data derived from the dataset and domain knowledge. This ensures that the system provides meaningful and practical outputs for real-world agricultural use.

4.4.1 Validation Methodology

4.4.2 Test Case Scenario

4.4.3 Results Comparison

4.4.4 Accuracy of Recommendations

4.4.5 Discussion

4.5 System Performance Evaluation

This section presents the evaluation of the overall system performance in terms of responsiveness, data processing efficiency, and reliability. The goal of this evaluation is to determine whether the developed system can operate effectively in real-world agricultural conditions and provide timely and consistent outputs to users.

The performance of the system was assessed using key indicators such as response time, data processing speed, and system stability during operation. The results provide insight into how well the system handles data flow from sensor acquisition to recommendation output.

4.5.1 Response Time Analysis

The response times for the Application Programming Interface (API) were measured through client-side network diagnostics to evaluate the latency of the TANIM system architecture. As detailed in **Table 9**, the diagnostic log captures the status, data size, and round-trip time for each primary endpoint utilized during a standard user session. These metrics provide a quantifiable baseline for assessing the efficiency of the backend infrastructure in delivering real-time agricultural data.

Dynamic requests, including the weather forecast, farm metadata, and

Table 9: TANIM System API Network Response and Latency Log

Endpoint Name	Status	Type	Size	Time
forecast?latitude=8.4542&longitude=...	200	fetch	0.8 kB	373 ms
farm?select=farm_id,farm_name,...	200	fetch	1.3 kB	388 ms
tanim_system?select=*	200	fetch	1.0 kB	512 ms
farmer?select=*	200	fetch	1.3 kB	509 ms
soil_health_test?select=*	200	fetch	1.3 kB	317 ms
farming_session?select=*&order=...	200	fetch	2.8 kB	325 ms
farming_session?farm_id=eq.8c	200	fetch	0.8 kB	325 ms
interactive_balloon.css	200	fetch	1.5 kB	8 ms
farming_session?select=*&order=...	200	fetch	2.8 kB	350 ms
farming_session?select=*&order=...	200	fetch	2.8 kB	289 ms

Note. Data retrieved from client-side browser network panel diagnostics.

active farming sessions, successfully returned a status code of 200 (OK). As shown in **Table 9**, the latency for these dynamic calls typically ranged from 289 ms to 512 ms. Specifically, the `tanim_system` and `farmer` endpoints recorded the highest latencies at 512 ms and 509 ms, respectively, while the `soil_health_test` endpoint responded in 317 ms. In contrast, the static resource `interactive_balloon.css` was retrieved in only 8 ms. This substantial difference highlights the expected overhead associated with server-side processing and database queries required for dynamic endpoints compared to static assets. Overall, the API performance remains well within the sub-second threshold required for responsive web applications.

4.5.2 Data Processing Performance

4.5.3 System Reliability

4.5.4 Discussion

4.6 Website Responsiveness and Accessibility

This section presents the evaluation of the website in terms of responsiveness and accessibility. The assessment focuses on how well the system adapts to different devices and how effectively it accommodates users with varying needs and abilities.

The website was evaluated across multiple screen sizes, including desktop, tablet, and mobile devices, to determine its consistency in layout, readability, and navigation. In addition, accessibility features such as text readability, color contrast, and basic keyboard navigation were examined to ensure that users can interact with the system without difficulty. The results provide insight into the system's ability to deliver a user-friendly and inclusive experience across different platforms.

4.6.1 Device Compatibility Testing

4.6.2 Responsiveness Evaluation

4.6.3 Accessibility Assessment

4.6.4 Performance Metrics

4.6.5 Discussion

4.7 System Testing

This section presents the results of the system testing conducted to evaluate the functionality, performance, and overall reliability of the developed system. The evaluation focuses on verifying whether the system meets the specified requirements and operates as intended.

The system was assessed using different testing methods, including functional testing, usability testing, compatibility testing, and performance testing. These methods were used to identify errors, measure system efficiency, and evaluate user interaction with the system. The results provide insight into the system's stability, responsiveness, and readiness for deployment.

4.7.1 Functional Testing

4.7.2 Integration Testing

4.7.3 Test Case Results

4.7.4 Error Handling and Robustness

4.7.5 Discussion

4.8 Usability Evaluation

This section presents a rigorous usability evaluation of the developed system, employing a structured, user-centered assessment framework to measure its effectiveness, efficiency, and user satisfaction across identified user groups. The evaluation was conducted using purposefully selected participants, consisting of a total of ten (10) respondents, specifically seven (7) farmers as primary end-users and three (3) Department of Agriculture (DA) officials as administrative users. This 7:3 distribution ensures adequate representation of both operational and managerial perspectives in system interaction.

A standardized instrument, the System Usability Scale (SUS), was utilized to obtain quantifiable measures of perceived usability, complemented by qualitative feedback to capture user experience nuances. The collected data were systematically analyzed to compute usability scores, interpret acceptability thresholds, and identify potential usability issues. This approach ensures that the evaluation is methodologically sound and aligned with established Human-Computer Interaction (HCI) practices, providing a reliable basis for assessing overall system usability and guiding subsequent design improvements.

4.8.1 Respondent Profile

4.8.2 Evaluation Instrument

4.8.3 Evaluation Criteria

4.8.4 Results and Analysis

4.8.5 Interpretation

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

This chapter provides the summary of the results obtained in this study and gives some recommendations for further investigation.

5.1 Summary of Findings

The study's findings address the initial research questions by confirming the effectiveness, reliability, and diverse applications of telemetry systems. The "Summary of Findings" section provides a concise overview of the key results from your research. This section should be factual and focus on presenting the data without interpretation. It should include:

Key Results:

Briefly summarize the most significant findings. Use bullet points or numbered lists for clarity if appropriate. Present the data as it was found, highlighting major patterns, relationships, or trends. Data Presentation:

Include tables, graphs, or charts that succinctly summarize the data. Make sure each visual aid is clearly labeled and includes a brief description.

Coverage of Research Questions:

Address each of the research questions or hypotheses posed at the be-

ginning of the study. Summarize the results relevant to each question.

5.2 Conclusion

The "Conclusions" section interprets the findings and discusses their implications. This section should:

Interpret Findings:

Provide an interpretation of the data summarized in the previous section. Discuss what the results mean in the context of the research questions or hypotheses. Implications:

Explain the significance of the findings. Discuss how the results contribute to the field of study or practical applications. Limitations:

Acknowledge any limitations in the study that may affect the results or their interpretation.

5.3 Recommendations

The "Recommendations" section provides actionable suggestions based on the study's findings and conclusions. This section should:

Practical Applications:

Offer specific recommendations for practitioners, policymakers, or other stakeholders based on the findings. Future Research:

Suggest areas for further investigation that could address the study's limitations or build on its findings. Implementation:

Provide guidance on how the recommendations can be implemented effectively.

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GANTT CHART



APPENDIX B

FUNCTIONAL TESTING FORM

A. Hardware and Data Capture

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	Sensor powers on and initializes correctly.	Sensor activates and displays readiness.		
2	The sensor captures soil parameters (N, P, K, pH, salinity, temperature, moisture).	Accurate readings displayed on serial monitor or app.		
3	Multiple readings under the same conditions are consistent.	Readings show minimal variance ($\pm 5\%$).		
4	ESP32 collects and processes sensor data properly.	Data packets successfully generated.		
5	The LoRa module transmits data to the gateway.	Data successfully received at gateway.		
6	Data stored correctly in the Firebase database.	Database shows complete and valid entries.		

B. Mobile Application (Farmer Side)

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	User login and authentication.	Users can log in using valid credentials.		
2	Soil health readings display correctly.	Dashboard shows latest sensor readings.		
3	Crop recommendation generation.	App suggests appropriate crops based on soil data.		
4	Fertilizer recommendation with dosage and schedule.	Correct fertilizer type and amount displayed.		
5	Multiple crop recommendations work without error.	The system handles multiple queries smoothly.		
6	Offline mode and data synchronization.	Data syncs automatically once reconnected.		
7	Farm information update.	Farmers can add/edit farm details successfully.		

C. Web Application (DA / Administrator Side)

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	Admin login and authentication.	DA officials can securely log in.		
2	Real-time soil health dashboard.	Data from sensors updates in real time.		
3	GIS mapping visualization.	Farm locations shown accurately on the map.		
4	Farmer profile management.	Officials can view and edit farmer data.		
5	Crop rules and recommendation management.	Admins can modify rules and save updates.		
6	Web responsiveness.	The system is accessible and functional on all devices.		

D. System-Wide Features

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	End-to-end data flow.	Data moves seamlessly from sensor → gateway → Firebase → apps.		
2	Error handling for missing/corrupt data.	System displays proper error messages or retries.		
3	Security and access control.	Only authorized users can access restricted features.		
4	Multi-user capability.	Multiple users can operate simultaneously without system crash.		

APPENDIX C

USABILITY TESTING SCALE

System Usability Scale (SUS) Questionnaire						
No.	Item	Frequency				
		SD	D	N	A	SA
1	I think that I would like to use this system frequently.					
2	I found the system unnecessarily complex.					
3	I think the system was easy to use.					
4	I think that I would need the support of a technical person to be able to use this system.					
5	I found the various functions in this system were well integrated.					
6	I thought there was too much inconsistency in this system.					
7	I would imagine that most people would learn to use this system very quickly.					
8	I found it difficult to navigate the system.					
9	I felt very confident using the system.					
10	I needed to learn a lot of things before I could get going with this system.					

APPENDIX D
TESTING CONDUCTED

APPENDIX E
LETTER OF CONSENT FOR INTERVIEW

APPENDIX F
LETTER OF REQUEST FOR SOIL TESTING KIT

APPENDIX G
DOCUMENTS REVIEWED

APPENDIX H
INTERVIEW QUESTIONS

APPENDIX I

CALIBRATION OF SOIL SENSOR USING SOIL TESTING KIT

APPENDIX J
DOCUMENTATION OF THE IMPLEMENTATION

APPENDIX K
FUNCTIONAL AND USABILITY TESTING CONDUCTED

APPENDIX L
SOIL LABORATORY RESULTS

APPENDIX M
SENSOR DATA SHEET

APPENDIX N
TURNITIN RESULTS

APPENDIX O
GRAMMARIAN RESULTS